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Spatio-temporal dynamic graph neural network-based missing measurement recovery method for power system state estimation

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HIGHLIGHTS

- Hybrid-missingness-aware STGNN models random and continuous missingness in power systems via joint node–edge–mask learning.
- Mask-enhanced dual-channel attention and dual-branch spatial framework strengthening node–edge collaboration under severe missingness.
- Bidirectional ODEs + Informer for cross-scale modeling of short-term fluctuations and long-range periodicity in real grids.

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ABSTRACT

High-quality measurement data are essential for reliable power system state estimation, yet SCADA and field monitoring devices frequently experience both random and continuous missingness due to sensor degradation, communication failures, and device outages. Although notable progress has been made in missing data recovery, many existing approaches remain tailored to low missing rates, underutilize node–edge electrical coupling, and rely on single-scale temporal modeling, which limits their robustness under the hybrid and high missingness conditions commonly observed in real grids. To address these limitations, this study proposes STGNI, a spatio-temporal graph neural network designed for heterogeneous missingness without predefined assumptions. The model integrates: (1) a missing-aware representation module that suppresses unreliable inputs; (2) a topology-informed dual-channel spatial module that captures coupled node–edge dependencies; and (3) a cross-scale temporal mechanism that jointly models short-term variations and long-range periodicity. Comprehensive experiments on the NREL-118 system and real provincial grid data show that STGNI consistently outperforms state-of-the-art baselines across 10–90% missing ratios, achieving substantial reductions in MAE and RMSE. The reconstructed measurements further enhance multiple state estimation algorithms, demonstrating the practical value of STGNI for improving data reliability and strengthening grid monitoring resilience.

1. Introduction

High-quality measurement data are fundamental to achieving accurate and reliable power system state estimation, which underpins real-time monitoring, secure operation, and decision-making in modern smart grids. As power networks continue to expand and increasingly integrate distributed energy resources, renewable generation, and diverse emerging loads, the operating conditions of power systems are becoming significantly more dynamic and uncertain. These trends amplify the need for accurate, real-time state awareness. However, in practical environments, measurements collected by Phasor Measurement Units (PMUs), Supervisory Control and Data Acquisition (SCADA) systems, and Advanced Metering Infrastructure (AMI) often suffer from various forms of missingness caused by equipment malfunctions, communication failures, and human errors [1]. Such missing data have been widely recognized as a major challenge in state estimation [2–4], as even moderate loss of measurements may lead to biased estimation, unstable system monitoring, or incorrect control actions.

In real-world grid operations, missingness typically manifests in two major forms: (1) random missingness, stemming from transient noise, packet loss, or temporary disturbances; and (2) block missingness, representing long continuous gaps caused by sustained device or communication failures. Sensor degradation, sustained outages, and communication congestion make such missing patterns persistent and

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difficult to avoid in practice. Unfortunately, many traditional imputation techniques implicitly assume random, independent missing entries, rendering them ineffective when both random and long-duration continuous missingness coexist.

1.1. Related work

Existing approaches for recovering missing power system data may be broadly classified into statistical, machine learning, and deep learning methods [5–8]. Statistical approaches—such as ARMA models [9,10], k-nearest neighbors (KNN) [11], and mean imputation [12,13]—are computationally efficient but insufficient for modeling nonlinear system dynamics or capturing spatiotemporal dependencies. Their performance deteriorates substantially under long gaps or large-scale missing segments [14,15].

Statistical methods—such as ARMA [9,10], KNN [11], and mean imputation [12,13]—are computationally efficient but unable to model nonlinear dynamics or spatiotemporal dependencies, leading to significant performance degradation under long gaps or large missing segments [14,15].

Machine-learning approaches offer greater flexibility but still face notable limitations. MDCTL-MCI [16] improves robustness via spectral decomposition and transfer learning, though its reliance on MSSA and limited covariates restricts its ability to represent complex spatiotemporal patterns. Fuzzy inductive reasoning [17] handles low-quality data through adaptive rules, but discretization limits high-dimensional temporal modeling. The 4CI model [18] tolerates high missing ratios using correlation-connected clusters, yet its lack of temporal modeling makes it unsuitable for PV-related time-series data. Consequently, ML-based approaches struggle with continuous missingness and fail to incorporate physical grid topology.

Deep-learning methods further strengthen nonlinear temporal modeling. Models such as Bi-LSTM–FONN [19] and Res-CNN-BiLSTM [20] improve wind or transformer data recovery but rely solely on temporal features, incur high computational cost, and do not model spatial correlations. Context-encoder methods like TCN-BiLSTM-CE [21] perform well on long gaps but lack explicit grid-topology awareness. Attention-based models such as SAITS [22] and BRITS [23] depend on iterative inference, making them vulnerable to error accumulation when consecutive gaps break temporal continuity. GAN-based imputers including CC-GAIN [24] improve distribution matching but remain prone to long-gap error propagation. Multi-stage pipelines like RF-GAIN [25] and ICEEMDAN–GAIN [26] increase accuracy through decomposition but introduce high complexity and still overlook grid topology. Collectively, these approaches emphasize temporal modeling but inadequately capture cross-node dependencies or multi-scale temporal dynamics.

To address spatial limitations, recent work adopts graph neural networks (GNNs), leveraging electrical-grid topology for reconstruction [27,28]. Representative methods include GAT-based aggregation [29], GCN–VAE frameworks [30], Gaussian-mixture-enhanced GATs [28], and multi-scale spatiotemporal GNNs [27]. While topology-aware, these models generally exhibit weak temporal modeling and struggle to jointly represent fast short-term disturbances and long-term periodicity. Moreover, evaluation under extremely high missing rates (80–90%)—common during communication outages or device failures—remains largely absent, leaving their robustness in real-world settings uncertain.

Overall, despite continuous progress, existing imputation methods still exhibit three major limitations: (1) Insufficient robustness under extremely high missing rates. Most models are evaluated only at moderate missing levels (≤ 40–50%), with little systematic assessment under the 80–90% long-duration missingness commonly observed in real grids. (2) Inadequate multi-scale temporal modeling. Single-scale recurrent or convolutional encoders struggle to capture both short-term disturbances and long-term periodic trends, limiting their ability to recover long consecutive gaps. (3) Incomplete utilization of grid topology and node–edge coupling. Current GNN-based approaches primarily model node connectivity while overlooking edge attributes, diffusion scales, and electrical coupling constraints, resulting in weak structural guidance under severe data loss.

These limitations highlight the need for a unified spatiotemporal imputation framework that integrates grid topology, multi-scale temporal structures, and missing-pattern awareness to maintain stable and reliable reconstruction under extreme missing conditions.

1.2. Contributions

To address this challenge, this paper proposes a spatio-temporal dynamic graph imputation model that integrates missing-pattern awareness, spatial structure modeling, and joint short- and long-term temporal modeling. In the spatial dimension, the model leverages the physical topology of the grid to achieve collaborative node–edge modeling; in the temporal dimension, it simultaneously captures short-term disturbances and long-term periodic patterns. With this design, the model significantly improves the accuracy and robustness of state recovery under complex missingness patterns.

- A unified imputation framework that jointly models node states, line attributes, and missing masks, enabling consistent representation of both random and long-duration missing patterns and providing stable performance under high missing ratios.
- A topology-aware spatial learning mechanism integrating multi-scale heat-kernel diffusion with edge-guided attention to capture node–edge electrical coupling and enhance robustness under topology perturbations and incomplete line information.
- A cross-scale temporal modeling framework is developed, integrating bidirectional neural ordinary differential equations to capture continuous short-term fluctuations and an Informer-style long-term periodicity encoder, thereby enabling accurate missing-data reconstruction across different temporal scales.
- Extensive evaluations on the NREL 118-bus system and real provincial grid data demonstrate that the proposed approach consistently outperforms state-of-the-art baselines under 10%–90% random missingness and long-duration continuous gaps, and significantly enhances the accuracy of downstream state estimation tasks.

The remainder of this paper is organized as follows. Section 3 introduces the fundamental definitions of graph structures and the main types of measurement missingness. Section 4 presents the overall framework and key modules of the proposed model in detail. Section 5 validates and analyzes the effectiveness of the proposed method through case studies. Section 5 concludes the paper and outlines potential directions for future research.

2. Definition

At each time step, the power system is abstracted as an undirected graph:

$$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathbf{X}_t^v, \mathbf{X}_t^e), \tag{1}$$

where $\mathcal{V}_t = \{1, 2, \ldots, N_v^{(t)}\}$ denotes the set of nodes corresponding to buses present at time step t; $\mathcal{E}_t \subseteq \mathcal{V}_t \times \mathcal{V}_t$ denotes the time-varying set of edges at time step t, corresponding to transmission lines, i.e., both nodes and edges may appear or disappear over time. $\mathbf{X}_t^v \in \mathbb{R}^{N_v^{(t)} \times d_v}$ is the node feature matrix, in which the i-th row $\mathbf{x}_{t,i} \in \mathbb{R}^{d_v}$ represents the measurement features of node i at time step t (e.g., voltage magnitude, power injection); and $\mathbf{X}_t^e \in \mathbb{R}^{N_e^{(t)} \times d_e}$ is the edge feature matrix with $N_e^{(t)} = |\mathcal{E}_t|$, in which each row corresponds to an existing edge $(i, j) \in \mathcal{E}_t$ at time step t, and stores its measurement features (e.g., active power at sending/receiving ends P_{linef}, P_{linet} and reactive power Q_{linef}, Q_{linet}).

To explicitly capture topology dynamics, we define the time-dependent adjacency indicator $A_t(i,j)$ as:

$$A_t(i,j) = \begin{cases} 1, & \text{if } (i,j) \in \mathcal{E}_t, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

This indicator distinguishes structural absence due to topology change (i.e., $(i,j) \notin \mathcal{E}_t$) from missing measurements on an existing line (i.e., $(i,j) \in \mathcal{E}_t$ but some entries of $\mathbf{X}_t^e$ are unobserved).

As the system evolves over time, a sequence of dynamic graphs is formed:

$$\mathcal{G}_{1:T} = \{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_T\}, \text{ where } \mathcal{V}_t \text{ and } \mathcal{E}_t \text{vary with } t. \tag{3}$$

In practical operation, measurement data at generator buses, including voltage magnitude, phase angle, and active/reactive power, are generally acquired through local monitoring devices with standard sensing and communication capabilities. Consequently, the probability of missing data at generator buses is extremely low. In contrast, transmission line measurements rely on remote devices installed at both ends of the line (e.g., SCADA terminals), whose communication links are more vulnerable to interruptions, sensor failures, or scheduled maintenance. Furthermore, long-distance transmission lines are subject to adverse weather, external disturbances, and equipment aging, which may lead to continuous data outages.

Based on these considerations, this work focuses on the imputation of transmission line power features (active/reactive) in the missing-data scenario, as this setting better reflects the primary source of data loss in real-world power grids and enhances the applicability of the proposed method. While the topology and node/edge sets may vary over time, the learning target in this work is the recovery of edge (line-flow) measurements; node features are used as contextual inputs when the corresponding nodes exist at time step t.

To this end, the edge-feature missing mask matrix defined only on existing edges at time step t $\mathbf{M}_t^e \in \{0,1\}^{N_e^{(t)}} \times d_e$ is defined as:

$$M^e_{t,(i,j),k} = \begin{cases} 1, & \text{if the k-th feature of edge } (i,j) \text{ at time step } t \text{ is missing,} \\ 0, & \text{otherwise,} \end{cases} \tag{4}$$

where the index (i,j) is valid only when $(i,j) \in \mathcal{E}_t$. For a specific edge (i,j), we denote its local mask vector as $\mathbf{m}_{t,(i,j)} = \mathbf{M}^e_{t,(i,j),:}$, which indicates the missing status of all features on that edge at time step t. The local mask vector can later be embedded through a mask encoder to enhance the model's awareness of missing patterns.

In mini-batch learning, topology dynamics lead to variable edge counts across time steps (i.e., $N_e^{(t)}$ is not constant). To enable temporal encoding over edge sequences, edge representations within each window/batch are padded to the maximum edge count in that batch, and an edge-validity mask is used so that temporal operators and attention/aggregation are applied only to edges that truly exist at each time step, while padded positions are ignored.

The missing patterns can be categorized into two types, i.e., random missingness and continuous missingness, as described in Section introduction. Formally, given the dynamic graph sequence $\{\mathbf{X}_t^v, \mathbf{X}_t^e, \mathbf{M}_t^e\}_{t=1}^T$, where edge features may be missing as indicated by the mask $\mathbf{M}_t^e \in \{0,1\}^{N_e^{(t)}} \times d_e$, the objective is to learn an imputation mapping:

$$f_\theta : \{\mathbf{X}_t^v, \tilde{\mathbf{X}}_t^e, \mathbf{M}_t^e\}_{t=1}^T \longrightarrow \{\hat{\mathbf{X}}_t^e\}_{t=1}^T, \tag{5}$$

where $\tilde{\mathbf{X}}_t^e \in \mathbb{R}^{N_e^{(t)}} \times d_e$ denotes the masked edge feature matrix obtained by replacing the missing entries in $\mathbf{X}_t^e$ with zeros according to $\mathbf{M}_t^e$. This zero-masked representation preserves the original feature dimensionality and provides a consistent input to the encoder, preventing the model from interpreting missing values as valid measurements. In contrast, $\hat{\mathbf{X}}_t^e$ denotes the reconstructed complete edge features predicted by the model. For training and evaluation under dynamic topology, the loss and metrics are computed only on entries that correspond to edges that exist at time step t (i.e., $(i,j) \in \mathcal{E}_t$) and feature positions marked as missing by $\mathbf{M}_t^e$, ensuring fair supervision when nodes/edges appear or disappear over time.

3. Model

The proposed model explicitly incorporates missing patterns through a missing-aware module, which encodes missing information at the feature level. At the relational level, a mask-enhanced dual-channel attention module jointly captures correlations of valid features and correlations implied by missing patterns. Furthermore, a dual-channel spatial structure modeling module integrates multi-scale heat kernel diffusion and edge-guided attention to jointly model the spatial topology and interaction strength of the power grid. Finally, a hybrid temporal modeling module fuses short-term dynamics with long-term dependencies to capture temporal evolution patterns of power system states, while residual-based prediction ensures physically consistent imputation of missing values.

Through this multi-level design, the model is able to fully exploit missing-pattern information, thereby significantly improving imputation accuracy and robustness under complex missingness patterns. The overall architecture of the STGNI model is illustrated in Fig. 1. Table 1 summarizes the inputs, outputs, and key computations of each module.

3.1. Missing-aware module

Power system measurements inherently exhibit strong structural and spatiotemporal correlations due to the grid topology and physical laws. When data are missing over consecutive time steps, these correlations can be leveraged to infer the unobserved values. However, conventional graph neural networks assume complete inputs and lack mechanisms to explicitly model missingness, limiting their robustness under incomplete data.

To enhance resilience against unreliable or missing measurements, we design a missing-aware gating mechanism that adaptively regulates message passing between nodes. Rather than directly imputing missing entries, the proposed module introduces the missing mask as an additional feature channel to quantify the reliability of each edge measurement. This enables the model to dynamically adjust how much information each edge contributes during aggregation. In other words, node features themselves are not corrected; instead, their updates are adaptively modulated according to the confidence of connected edges.

This design serves two purposes: (i) it suppresses the propagation of errors from unreliable or missing edges that could otherwise distort node representations, and (ii) it exploits structured missingness patterns to implicitly capture potential topological variations, improving generalization across diverse grid conditions.

The computation of this module proceeds as follows. First, node features, edge features, and mask features are mapped into a shared hidden space:

$$\mathbf{h}_x = \text{LeakyReLU}\left(\mathbf{W}_x \cdot \mathbf{x}_i\right), \tag{6}$$

$$\mathbf{h}_{e_{ij}} = \text{LeakyReLU}\left(\mathbf{W}_e \cdot \mathbf{e}_{ij}\right), \tag{7}$$

$$\mathbf{h}_m = \text{LeakyReLU}\left(\mathbf{W}_m \cdot (\mathbf{1} - \mathbf{m})\right), \tag{8}$$

where $\mathbf{x}_i$ denotes the node features, $\mathbf{e}_{ij}$ the features of edge (i,j), and $\mathbf{m} = \mathbf{m}_{t,(i,j)} \in \{0,1\}^{d_m}$ is the local edge-wise missing mask (1 indicates missing), and $\mathbf{W}_x$, $\mathbf{W}_e$, $\mathbf{W}_m$ are the corresponding projection matrices. Next, a gating factor is generated based on the mask information:

$$\mathbf{g} = \sigma\left(\mathbf{h}_m\right), \tag{9}$$

Here, $\mathbf{g}$ is a feature-wise reliability gate derived solely from the missing mask (through the availability indicator $\mathbf{1} - \mathbf{m}$): it indicates how

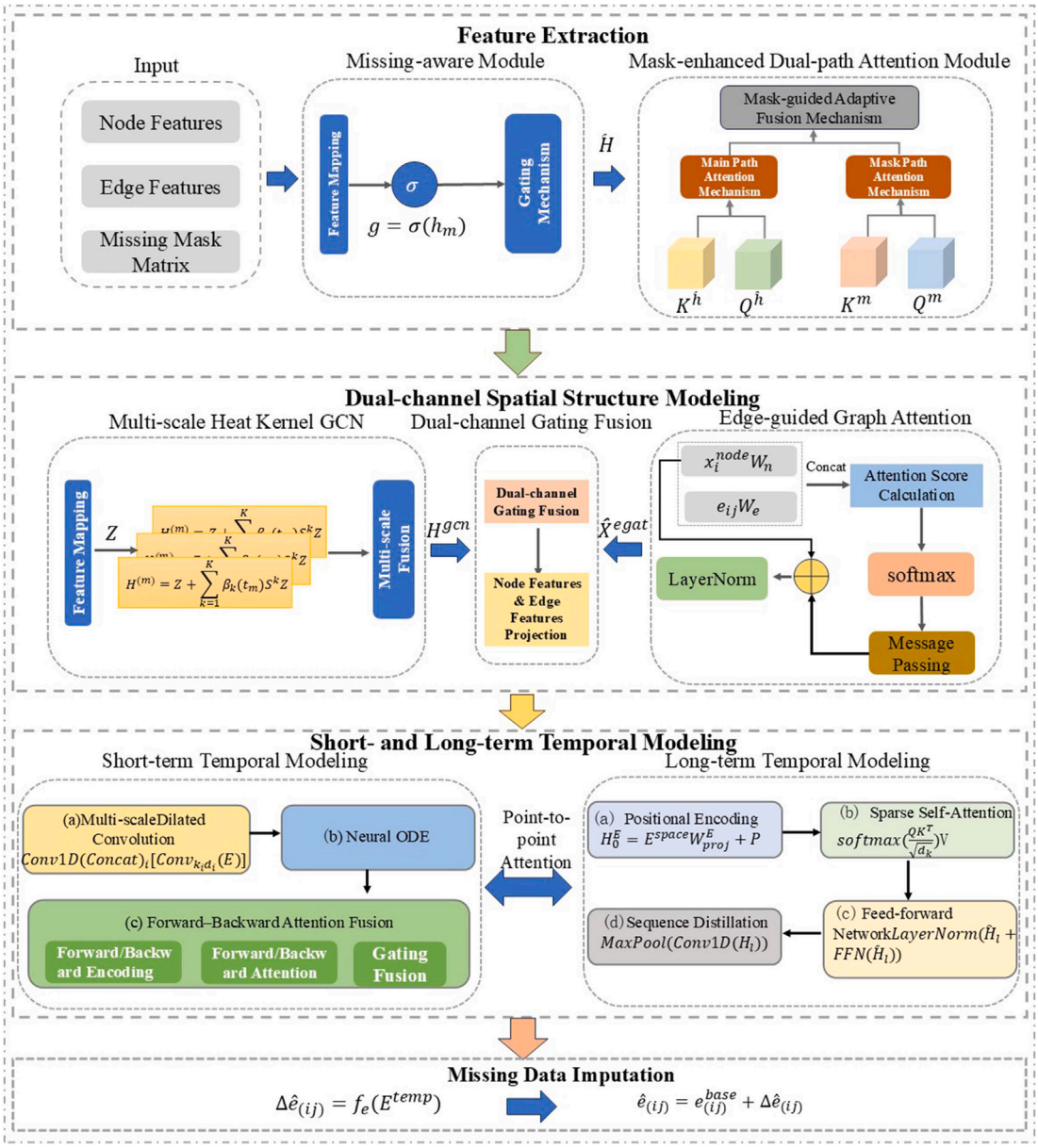


Fig. 1. Framework of the STGNI Model.

much each edge feature dimension should be trusted at the current time step. When an edge measurement is missing or unreliable in certain dimensions, the corresponding entries in $\mathbf{g}$ tend to be smaller, thereby suppressing the injection of $\mathbf{h}_{e_{ij}}$ into the fused representation; when the measurement is available and reliable, $\mathbf{g}$ tends to be larger and allows edge information to contribute more strongly.

This gating factor is used to modulate the edge features before they are fused with node features:

$$\mathbf{h}_{\text{fused}} = \mathbf{h}_x + \mathbf{g} \odot \mathbf{h}_{e_{ij}}, \tag{10}$$

where $\odot$ denotes the Hadamard (element-wise) product.

Finally, Layer Normalization and Dropout are applied to improve training stability and generalization:

$$\tilde{\mathbf{h}} = \text{LayerNorm}\left(\text{Dropout}\left(\mathbf{h}_{\text{fused}}\right)\right). \tag{11}$$

Note that $\tilde{\mathbf{h}}$ is computed for each edge (i, j) (i.e., edge-indexed), since it fuses the edge attribute and the mask of that edge with the node feature projection. The resulting edge-level representations are denoted as $\tilde{\mathbf{H}}^e = \left[\tilde{\mathbf{h}}_{(i,j)}\right]_{(i,j)\in\mathcal{E}} \in \mathbb{R}^{N_e \times d_h}$. To interface with subsequent *node-based* spatial modules, we transform $\tilde{\mathbf{H}}^e$ into node-level representations via edge-to-node aggregation over incident edges. For node v, let $\mathcal{E}_v = \{(u, v) \in \mathcal{E}\} \cup \{(v, u) \in \mathcal{E}\}$ denote its incident edge set. We compute:

$$\mathbf{x}_v^{\text{ma}} = \text{Agg}\left(\left\{\tilde{\mathbf{h}}_e : e \in \mathcal{E}_v\right\}\right) = \frac{1}{|\mathcal{E}_v|} \sum_{e\in\mathcal{E}_v} \tilde{\mathbf{h}}_e, \tag{12}$$

where $\mathbf{X}^{\text{ma}} = \left[\mathbf{x}_v^{\text{ma}}\right]_{v\in\mathcal{V}} \in \mathbb{R}^{N_v \times d_h}$. This missing-aware module explicitly leverages missing-pattern information and demonstrates robustness to both random and continuous missing cases, with particularly strong performance in structured missingness.

3.2. Mask-enhanced dual-channel attention module

In power grids, missing information not only manifests as absent feature values but also perturbs correlation patterns among measurements. Under structured missingness (e.g., consecutive gaps) or large-scale

Table 1
Summary of the proposed STGNI model: inputs, outputs, and key computations of each module.

Module	Input	Output	Key computation (core operators)
Missing-Aware Module (Section 3.1)	Node feat. $\mathbf{x}_i$; Edge feat. $\mathbf{e}_{ij}$; Edge mask $\mathbf{m}_{t,(i,j)}$	Node-level missing-aware feat. $\mathbf{X}^{\text{ma}} \in \mathbb{R}^{N_e \times d_h}$	(i) Projections: $\mathbf{h}_x = \phi(\mathbf{W}_x \mathbf{x}_i)$, $\mathbf{h}_e = \phi(\mathbf{W}_e \mathbf{e}_{ij})$, $\mathbf{h}_m = \phi(\mathbf{W}_m(\mathbf{1} - \mathbf{m}))$; (ii) Reliability gate: $\mathbf{g}^{\text{ma}} = \sigma(\mathbf{h}_m)$; (iii) Mask-aware fusion: $\mathbf{h}_{\text{fused}} = \mathbf{h}_x + \mathbf{g}^{\text{ma}} \odot \mathbf{h}_e$; (iv) Edge-to-node aggregation: $\mathbf{X}^{\text{ma}} = \text{Agg}_{e \to v}(\tilde{\mathbf{H}}^e)$.
Mask-Enhanced Dual-Channel Attention (Section 3.2)	Edge repr. $\tilde{\mathbf{H}}^e$; Mask $\mathbf{M}$	Node-level attention feat. $\mathbf{X}^{\text{attn}} \in \mathbb{R}^{N_e \times d_h}$	(i) Main-path attention: $\mathbf{S}_{\text{main}} = \text{Softmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d})$, $\mathbf{X}_{\text{main}} = \mathbf{S}_{\text{main}}\tilde{\mathbf{H}}^e$; (ii) Mask-path attention: $\mathbf{S}_{\text{mask}} = \text{Softmax}(\mathbf{Q}\mathbf{K}_m^\top/\sqrt{d})$, $\mathbf{X}_{\text{mask}} = \mathbf{S}_{\text{mask}}\tilde{\mathbf{H}}^e$; (iii) Mask-guided gate: $\boldsymbol{\alpha} = \sigma(\mathbf{M}\mathbf{W}_\alpha)$; (iv) Fusion + norm: $\mathbf{H}^{e,\text{attn}} = \text{LN}(\mathbf{X}_{\text{main}} + \boldsymbol{\alpha} \odot \mathbf{X}_{\text{mask}})$; (v) Edge-to-node aggregation: $\mathbf{X}^{\text{attn}} = \text{Agg}_{e \to v}(\mathbf{H}^{e,\text{attn}})$.
Dual-Channel Spatial Structure Modeling (Section 3.3)	Node input $\mathbf{X}^{\text{attn}}$; Edge attr. $\mathbf{e}_{ij}$	Spatially refined edge feat. $\mathbf{E}^{\text{space}}$	(i) GCN branch (multi-scale heat diffusion): $\mathbf{H}^{(m)} = \mathbf{Z} + \sum_{k=1}^{K} \beta_k(t_m)\mathbf{S}^k\mathbf{Z}$, $\mathbf{H}^{\text{gcn}} = \sum_{m=1}^{M} \omega_m \mathbf{H}^{(m)}$; (ii) EGAT branch: edge-guided attention α_{ij} + message passing with $\mathbf{e}_{ij}$; (iii) Dual-branch gated fusion: $\mathbf{g}^{\text{sp}} = \sigma([\mathbf{H}^{\text{gcn}} \| \hat{\mathbf{X}}^{\text{egat}}]\mathbf{W}_g^{\text{sp}} + \mathbf{b}_g^{\text{sp}})$, $\mathbf{X}^{\text{space}} = \mathbf{g}^{\text{sp}} \odot \mathbf{H}^{\text{gcn}} + (1 - \mathbf{g}^{\text{sp}}) \odot \hat{\mathbf{X}}^{\text{egat}}$; (iv) Node-to-edge restoration: $[\mathbf{X}_i^{\text{space}} \| \mathbf{X}_j^{\text{space}}] \mapsto \mathbf{e}_{(i,j)}^{\text{space}}$.
Short-Term Temporal Encoder (BiODE) (Section 3.4)	Sequence $\mathbf{E}^{\text{space}}$ ($T \times N_e \times d_h$)	Short-term repr. $\mathbf{E}^{\text{short}}$	(i) IncepTCN: multi-scale dilated Conv1D over time; (ii) Neural ODE solver (`odeint`, steps S) for continuous- time dynamics; (iii) Forward–backward attention fusion (multi-head) to combine $\mathbf{z}_{\text{fwd}}$ and $\mathbf{z}_{\text{bwd}}$.
Long-Term Temporal Encoder (Informer) (Section 3.4)	Sequence $\mathbf{E}^{\text{space}}$ ($T \times N_e \times d_h$)	Long-term repr. $\mathbf{E}^{\text{long}}$	(i) ProbSparse attention (Top-u queries) for long sequences; (ii) Position-wise FFN; (iii) Multi-head linear projections for $Q/K/V$ and output.
Short–Long Fusion (Section 3.4.1)	$\mathbf{E}^{\text{short}}$, $\mathbf{E}^{\text{long}}$	Temporal repr. $\mathbf{E}^{\text{temp}}$	Point-to-point cross-scale attention (edge-wise alignment) + residual connection + normalization to fuse short/long representations.
Residual Imputation Predictor (Section 3.4.2)	Temporal repr. $\mathbf{E}^{\text{temp}}$; Base meas. $\mathbf{e}^{\text{base}}$	Imputed edge $\hat{\mathbf{e}}_{(i,j)}$	Residual prediction: $\Delta\hat{\mathbf{e}}_{(i,j)} = f_e(\mathbf{E}^{\text{temp}})$; Final imputation: $\hat{\mathbf{e}} = \mathbf{e}^{\text{base}} + \Delta\hat{\mathbf{e}}$.

random missingness, correlations among valid measurements and correlations induced by missing patterns coexist and are often coupled. If attention is computed solely from observed features, relational cues implied by missingness may be overlooked.

To address this issue, we propose a mask-enhanced dual-channel attention mechanism, consisting of (i) a main-path attention focusing on correlations among edge representations, and (ii) a mask-path attention capturing correlations implied by missing patterns. The two paths are fused through a mask-guided adaptive gate, enabling robust exploitation of reliable information while suppressing unreliable signals.

Edge-level inputs. Let $\tilde{\mathbf{H}}^e \in \mathbb{R}^{N_e \times d_h}$ denote the edge-level representations produced by the Missing-Aware Module, and let $\mathbf{M} \in \mathbb{R}^{N_e \times d_m}$ denote the edge-wise missing mask vectors (1 indicates missing). We first embed mask features into the hidden space:

$$\mathbf{H}_m = \text{LeakyReLU}\left(\mathbf{M}\mathbf{W}_m^{(a)}\right) \in \mathbb{R}^{N_e \times d_h}. \tag{13}$$

Then we form query/key projections:

$$\mathbf{Q} = \tilde{\mathbf{H}}^e\mathbf{W}_Q, \quad \mathbf{K} = \tilde{\mathbf{H}}^e\mathbf{W}_K, \quad \mathbf{K}_m = \mathbf{H}_m\mathbf{W}_{K_m}, \tag{14}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_{K_m} \in \mathbb{R}^{d_h \times d}$ and $\mathbf{Q}, \mathbf{K}, \mathbf{K}_m \in \mathbb{R}^{N_e \times d}$.

(1) Main-Path Attention (edge–edge correlations).

$$\mathbf{S}_{\text{main}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad \mathbf{X}_{\text{main}} = \mathbf{S}_{\text{main}}\tilde{\mathbf{H}}^e, \tag{15}$$

where $\mathbf{S}_{\text{main}} \in \mathbb{R}^{N_e \times N_e}$ and $\mathbf{X}_{\text{main}} \in \mathbb{R}^{N_e \times d_h}$.

(2) Mask-Path Attention (missing-pattern correlations).

$$\mathbf{S}_{\text{mask}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}_m^\top}{\sqrt{d}}\right), \quad \mathbf{X}_{\text{mask}} = \mathbf{S}_{\text{mask}}\tilde{\mathbf{H}}^e, \tag{16}$$

where $\mathbf{S}_{\text{mask}} \in \mathbb{R}^{N_e \times N_e}$ and $\mathbf{X}_{\text{mask}} \in \mathbb{R}^{N_e \times d_h}$. Note that the mask path uses $\mathbf{H}_m$ only to derive attention weights (i.e., relational cues implied by missing patterns), while the values remain $\tilde{\mathbf{H}}^e$ to propagate measurement-relevant representations.

(3) Mask-Guided Adaptive Fusion. We generate a feature-wise adaptive gate from the mask vectors:

$$\boldsymbol{\alpha} = \sigma\left(\mathbf{M}\mathbf{W}_\alpha\right) \in \mathbb{R}^{N_e \times d_h}, \tag{17}$$

where $\mathbf{W}_\alpha \in \mathbb{R}^{d_m \times d_h}$. Intuitively, $\boldsymbol{\alpha}$ is learned from the edge-wise mask patterns to adaptively balance the main-path and mask-path attentions: it increases the contribution of $\mathbf{X}_{\text{mask}}$ when mask-induced relational cues are beneficial, while avoiding over-reliance when observed measurements already provide sufficient correlations.

The fused edge-level attention output is:

$$\mathbf{H}^{e,\text{attn}} = \text{LayerNorm}\left(\mathbf{X}_{\text{main}} + \boldsymbol{\alpha} \odot \mathbf{X}_{\text{mask}}\right) \in \mathbb{R}^{N_e \times d_h}. \tag{18}$$

In practice, when N_e is large, the attention computation can be implemented in a sparse or top-k manner to reduce the $\mathcal{O}(N_e^2)$ cost without changing the formulation.

Edge-to-node aggregation for spatial modeling. Since the subsequent spatial module is node-indexed, we aggregate $\mathbf{H}^{e,\text{attn}}$ from edges to nodes using incident edges:

$$\mathbf{x}_v^{\text{attn}} = \text{Agg}_{\mathcal{E}\to v}\left(\left\{\mathbf{H}_e^{e,\text{attn}} : e \in \mathcal{E}_v\right\}\right) = \frac{1}{|\mathcal{E}_v|}\sum_{e\in\mathcal{E}_v}\mathbf{H}_e^{e,\text{attn}}, \tag{19}$$

yielding $\mathbf{X}^{\text{attn}} = \left[\mathbf{x}_v^{\text{attn}}\right]_{v\in\mathcal{V}} \in \mathbb{R}^{N_v\times d_h}$, which serves as the node-level input to the spatial structure modeling module.

Through the synergy between the missing-aware module and the mask-enhanced dual-channel attention module, the model can perceive missing patterns at both the feature and relational levels, thereby improving robustness under complex missing scenarios.

3.3. Dual-channel spatial structure modeling module

In the scenario of missing measurements in power grids, relying on a single spatial modeling mechanism exhibits evident limitations: GCN-based methods leverage topological diffusion to preserve global consistency but are highly sensitive to missing edges, making it difficult to restore disrupted interactions; GAT-based methods incorporate edge information but fail to explicitly exploit the physical topology of power grids, leading to insufficient generalization and robustness in complex networks.

To overcome these limitations, we propose an innovative *dual-channel spatial structure modeling module*, composed of a multi-scale heat kernel graph convolution and an edge-guided graph attention, combined via a gated adaptive fusion mechanism. The core innovations of this design are as follows: (i) introducing multi-scale heat kernel diffusion into the GCN branch, where learnable temperature parameters adaptively balance the contributions of neighbors at different orders, thereby enhancing topological consistency; (ii) explicitly incorporating edge features into attention computation and message passing in the EGAT branch, enabling compensation for missing-edge information; (iii) applying a gated adaptive mechanism during dual-channel fusion, allowing the model to dynamically adjust its reliance on the GCN or EGAT branch depending on missing patterns, and finally restoring node features back to edge space to achieve measurement completion. The node input to this module is $\mathbf{X}^{\text{attn}} \in \mathbb{R}^{N_v\times d_h}$. The edge attributes used by the EGAT branch are the missing-aware edge measurements (or their refined estimates in the second pass); for simplicity, we denote them as $\mathbf{e}_{ij}$ in the following derivations. This module ensures a balance between topological consistency and missing-information compensation, providing robust spatial feature representations for subsequent temporal modeling and imputation.

(1) Multi-Scale Heat Kernel Graph Convolution Given $\mathbf{X}^{\text{attn}} \in \mathbb{R}^{N_v\times d_h}$, a linear projection is first applied:

$$\mathbf{Z} = \mathbf{X}^{\text{attn}} \cdot \mathbf{W}, \tag{20}$$

followed by multi-order heat kernel diffusion:

$$\mathbf{H}^{(m)} = \mathbf{Z} + \sum_{k=1}^{K}\beta_k(t_m)\mathbf{S}^k\mathbf{Z}, \quad m = 1,\dots,M, \tag{21}$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with self-loops, $\mathbf{I}$ denotes the identity matrix, $\tilde{\mathbf{D}}$ is the corresponding degree matrix, and $\tilde{\mathbf{S}} = \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$ is the normalized adjacency matrix. $\mathbf{S}^k$ denotes k-hop diffusion. t_m is the learnable temperature parameter of the m-th scale, controlling diffusion decay speed. $\beta_k(t) = \frac{e^{-tk}(tk)^k}{k!}$ is the heat kernel coefficient. K is the approximation order and M the number of scales. The multi-scale fusion is computed as:

$$\mathbf{H}^{\text{gcn}} = \sum_{m=1}^{M}\omega_m\mathbf{H}^{(m)}, \quad \sum_{m=1}^{M}\omega_m = 1, \tag{22}$$

where ω_m denotes learnable fusion weights.

(2) Edge-Guided Graph Attention Node and edge features are first mapped to a unified dimension:

$$\mathbf{x}_i' = \mathbf{x}_i^{\text{node}}\mathbf{W}_n, \quad \mathbf{e}_{ij}' = \mathbf{e}_{ij}\mathbf{W}_e, \tag{23}$$

The attention score is then computed as:

$$s_{ij} = \mathbf{a}^\top[\mathbf{x}_i'\|\mathbf{x}_j'\|\mathbf{e}_{ij}'], \tag{24}$$

and normalized as:

$$\alpha_{ij} = \frac{\exp(s_{ij})}{\sum_{k\in\mathcal{N}(i)}\exp(s_{ik})}, \tag{25}$$

Message passing and node update are performed as:

$$\mathbf{m}_{ij} = \mathbf{W}_m[\mathbf{x}_j'\|\mathbf{e}_{ij}'], \tag{26}$$

$$\hat{\mathbf{x}}_i^{\text{egat}} = \text{LayerNorm}\left(\sum_{j\in\mathcal{N}(i)}\alpha_{ij}\mathbf{m}_{ij} + \mathbf{x}_i'\right), \tag{27}$$

where $[\cdot\|\cdot]$ denotes feature concatenation, and LayerNorm is applied for feature-wise normalization.

(3) Dual-Channel Gated Fusion We combine the GCN and EGAT branches using a node-wise, feature-wise fusion gate. Specifically, the gate is generated from the concatenated branch outputs:

$$\mathbf{g}^{\text{sp}} = \sigma\left(\left[\mathbf{H}^{\text{gcn}}\,\|\,\hat{\mathbf{X}}^{\text{egat}}\right]\mathbf{W}_g^{\text{sp}} + \mathbf{b}_g^{\text{sp}}\right), \tag{28}$$

and the fused node representation is computed as

$$\mathbf{X}^{\text{space}} = \mathbf{g}^{\text{sp}}\odot\mathbf{H}^{\text{gcn}} + \left(\mathbf{1} - \mathbf{g}^{\text{sp}}\right)\odot\hat{\mathbf{X}}^{\text{egat}}. \tag{29}$$

Here, $\mathbf{W}_g^{\text{sp}}$ and $\mathbf{b}_g^{\text{sp}}$ are learnable parameters of the fusion gate, implemented as a single linear layer followed by a Sigmoid nonlinearity. $\mathbf{g}^{\text{sp}} \in (0,1)^{N_v\times d_h}$, $\sigma(\cdot)$ denotes the Sigmoid function, and $\odot$ denotes element-wise multiplication. A larger gate value increases the contribution of the topology-preserving GCN branch, whereas a smaller value favors the edge-attribute-aware EGAT branch.

(4) Node-to-Edge Restoration Finally, node features are mapped back to the edge feature space:

$$\mathbf{z}_{(i,j)} = [\mathbf{X}_i^{\text{space}}\|\mathbf{X}_j^{\text{space}}], \tag{30}$$

$$\mathbf{e}_{(i,j)}^{\text{space}} = \text{LeakyReLU}(\mathbf{W}_o\mathbf{z}_{(i,j)} + \mathbf{b}_o), \tag{31}$$

Here, N_v and N_e denote the number of nodes and edges, respectively, and d_h is the hidden dimension. $\mathcal{N}(i)$ denotes the neighbor set of node i, and $\mathcal{E}i^{\text{out}}$ its outgoing edge set. $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{S}}$ denote the adjacency matrix with self-loops and its normalized version, respectively. $\mathbf{S}^k$ is the k-hop diffusion matrix; t_m the temperature parameter; $\beta_k(t)$ the heat kernel coefficient; K the diffusion approximation order; M the number of scales; ω_m the multi-scale fusion weights; αij the local softmax attention weight; and $\mathbf{a}$, $\mathbf{W}_n$, $\mathbf{W}_e$, $\mathbf{W}_m$, $\mathbf{W}_g$, $\mathbf{W}_o$ are learnable parameter matrices.

3.4. Short–long term temporal modeling module

In the imputation of edge features in power grids, both short-term rapid fluctuations (e.g., disturbances, measurement noise) and long-term smooth evolutions (e.g., load curves, operational trends) coexist. Relying solely on a single temporal modeling mechanism may lead to the following issues: convolution-based methods can capture local dependencies but fail to represent continuous physical evolution over time; Ordinary Differential Equation-based(ODE-based) continuous modeling provides smooth trajectories but is sensitive to short-term abrupt changes; unidirectional encoding results in insufficient utilization of both past and future information.

(1) Bidirectional Neural Differential Temporal Encoder: To address the above limitations, this module employs multi-scale dilated convolutions to capture local dynamics, combines neural ODEs for smooth continuous-time modeling, and incorporates forward–backward attention fusion to jointly leverage historical and future information. Thus, the design is sensitive to short-term disturbances while preserving continuity and smoothness across time.

(a) Multi-Scale Dilated Convolution For edge features, multiple 1D convolutions with different kernel sizes k_i and dilation rates d_i are applied to model multi-scale local temporal dependencies:

$$\mathrm{Conv}_{k_i,d_i}(\mathbf{E}) = \mathrm{ReLU}\Big(\mathrm{BN}\Big(\mathrm{Conv1D}_{k_i,d_i}(\mathbf{E}^{\mathrm{space}})\Big)\Big), \tag{32}$$

where B is the batch size, C is the number of feature channels, and T is the sequence length. $\mathrm{Conv1D}_{k_i,d_i}$ denotes a 1D convolution with kernel size k_i and dilation rate d_i; BN denotes batch normalization, and ReLU is the activation function.

Outputs from different scales are concatenated along the channel dimension and then compressed using a 1×1 convolution:

$$\mathrm{IncepTCN}(\mathbf{E}) = \mathrm{Conv1D}_{1\times 1}\Big(\mathrm{Concat}_i\Big[\mathrm{Conv}_{k_i,d_i}(\mathbf{E})\Big]\Big), \tag{33}$$

where $\mathrm{Concat}_i[\cdot]$ denotes concatenation across different scales along the channel dimension.

(b) Neural Ordinary Differential Equation The output of $\mathrm{IncepTCN}(\mathbf{E})$ is used as the initial latent state $\mathbf{z}_0$. The continuous-time dynamics are then modeled by a neural ODE [31], as illustrated in Fig. 2

$$\frac{d\mathbf{z}(t)}{dt} = f_\theta(\mathbf{z}(t), t), \quad \mathbf{z}(0) = \mathbf{z}_0, \tag{34}$$

where $\mathbf{z}_0 \in \mathbb{R}^C$ represents the initial state of a node or edge at $t = 0$, t is the continuous time variable, and f_θ is parameterized by a two-layer MLP:

$$f_\theta(\mathbf{z}) = W_2 \cdot \tanh\big(W_1\mathbf{z} + b_1\big) + b_2, \tag{35}$$

where W_1, W_2 are weight matrices, b_1, b_2 are bias terms, and $\tanh(\cdot)$ is the hyperbolic tangent activation function.

The temporal evolution is solved using the Dormand–Prince (DOPRI5) adaptive Runge–Kutta integrator implemented in $\mathrm{odeint}(\cdot)$:

$$\mathbf{z}(t) = \mathrm{odeint}\big(f_\theta, \mathbf{z}_0, t\big), \tag{36}$$

yielding smooth and physically consistent continuous-time representations.

(c) Forward–Backward Attention Fusion The temporal sequence is encoded in both forward and backward directions, producing $\mathbf{z}_{\mathrm{fwd}}$ and $\mathbf{z}_{\mathrm{bwd}}$, which are fused through an attention mechanism:

$$\mathbf{Q} = \mathbf{W}_q\mathbf{z}_{\mathrm{fwd}}, \quad \mathbf{K} = \mathbf{W}_k\mathbf{z}_{\mathrm{bwd}}, \quad \mathbf{V} = \mathbf{W}_v\mathbf{z}_{\mathrm{bwd}}, \tag{37}$$

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad \mathbf{O} = \mathbf{A}\cdot\mathbf{V}, \tag{38}$$

where d is the scaling factor (feature dimension), $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v$ are learnable projection matrices, and $\mathbf{A}$ is the attention weight matrix.

A gating mechanism balances the fusion:

$$\mathbf{g}^{\mathrm{tmp}} = \sigma\Big(\mathbf{W}_g^{\mathrm{tmp}}\left[\mathbf{z}_{\mathrm{fwd}}, \mathbf{z}_{\mathrm{bwd}}\right]\Big), \quad \mathbf{z}^{\mathrm{fused}} = \mathbf{g}^{\mathrm{tmp}} \odot \mathbf{O} + (1 - \mathbf{g}^{\mathrm{tmp}}) \odot \mathbf{z}_{\mathrm{fwd}}. \tag{39}$$

where $\sigma(\cdot)$ is the Sigmoid function, $\mathbf{W}_g$ is the fusion weight matrix, and $\odot$ denotes element-wise multiplication. The gate $\mathbf{g}^{\mathrm{tmp}}$ has the same shape as $\mathbf{z}_{\mathrm{fwd}}$ (broadcast if needed), enabling feature-wise fusion. Notably, $\mathbf{O}$ serves as a context-dependent refinement of $\mathbf{z}_{\mathrm{fwd}}$ using $\mathbf{z}_{\mathrm{bwd}}$, whereas $\mathbf{g}^{\mathrm{tmp}}$ determines the refinement strength rather than enforcing it uniformly.

Finally, the short-term temporal representation is obtained:

$$\mathbf{E}^{\mathrm{short}} = \mathrm{BiODE}(\mathbf{E}^{\mathrm{space}}), \tag{40}$$

where BiODE denotes our Bidirectional Neural ODE temporal encoder which integrates multi-scale convolution, continuous-time ODE dynamics, and forward–backward attention fusion.

(2) Informer-Based Long-Term Temporal Encoder: To capture periodicity and long-term trends of edge features, we introduce the Informer [32] with its sparse self-attention mechanism, adapted for the edge feature imputation task. Compared with standard self-attention, the ProbSparse strategy in Informer effectively alleviates computational redundancy in long-sequence modeling. We embed this encoder into the imputation framework, complementing the short-term encoder to simultaneously capture local dynamics and long-term trends. This short–long collaborative design mitigates the temporal distortion and prediction bias often observed in single-scale methods.

(a) Input Mapping and Positional Encoding For edge features $\mathbf{E}^{\mathrm{space}}$, a linear projection maps them into a d_{model}-dimensional space, followed by positional encoding $\mathbf{P}$:

$$\mathbf{H}_0^E = \mathbf{E}^{\mathrm{space}}\mathrm{W}_{\mathrm{proj}}^E + \mathbf{P}, \tag{41}$$

The resulting $\mathbf{H}_0^E$ is fed into the first encoder layer.

(b) Sparse Self-Attention In the l-th encoder layer, sparse self-attention reduces redundant computations by retaining only query vectors with significant contributions to the attention scores. The formulation is:

$$\mathbf{Z}_l = \mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{42}$$

where $\mathbf{Q} = \mathbf{H}_{l-1}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{H}_{l-1}\mathbf{W}_K$, $\mathbf{V} = \mathbf{H}_{l-1}\mathbf{W}_V$, and d_k is the key dimension. ProbSparse selects the Top-u queries ($u \ll L$) to reduce redundant computations.

(c) Feed-Forward Network The attention output passes through residual connections and normalization before entering a position-wise feed-forward network:

$$\tilde{\mathbf{H}}_l = \mathrm{LayerNorm}\left(\mathbf{H}_{l-1} + \mathbf{Z}_l\right), \quad \mathbf{H}_l = \mathrm{LayerNorm}\left(\tilde{\mathbf{H}}_l + \mathrm{FFN}\left(\tilde{\mathbf{H}}_l\right)\right), \tag{43}$$

where W_1, W_2 are learnable parameters, and d_{ff} is the feed-forward dimension.

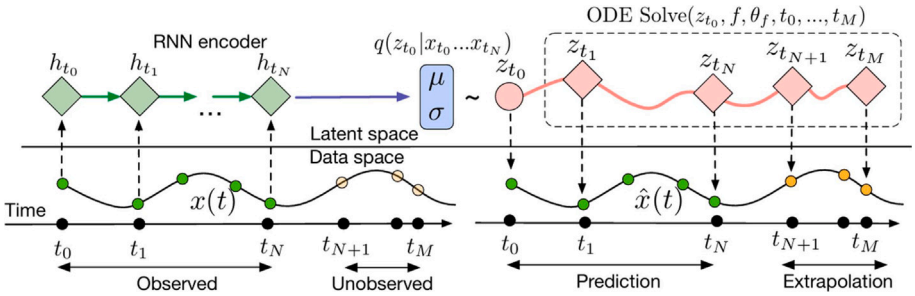


Fig. 2. Computation graph of the latent ODE model [31].

(d) Sequence Distillation Informer introduces convolutional downsampling and max pooling between layers to gradually compress sequence length:

$$\mathbf{H}_l^{\text{down}} = \text{MaxPool}\left(\text{Conv1D}\left(\mathbf{H}_l\right)\right), \tag{44}$$

(e) Long-Term Representation of Edge Features After L stacked encoder layers, the long-term dependency representation of edge features is obtained:

$$\mathbf{E}^{\text{long}} = \mathbf{H}_L, \tag{45}$$

3.4.1. Short–long fusion

To address the issues of computational redundancy, cross-edge interference, and misalignment between short- and long-term temporal modeling in power grids, we propose a cross-scale point-to-point attention mechanism. This mechanism fuses the short- and long-term representations of the same edge in a one-to-one manner, reducing computational complexity while ensuring temporal alignment and physical consistency. This significantly improves imputation accuracy and robustness under high missing rates and long gaps.

$$\mathbf{E}^{\text{temp}} = \text{Fuse}(\mathbf{E}^{\text{short}}, \mathbf{E}^{\text{long}}), \tag{46}$$

where $\text{Fuse}(\cdot)$ consists of point-to-point attention, residual connections, and normalization.

3.4.2. Missing value imputation

Finally, the residual of missing values is predicted:

$$\Delta\hat{\mathbf{e}}_{(i,j)} = f_e(\mathbf{E}^{\text{temp}}), \tag{47}$$

where $f_e(\cdot)$ is an MLP predictor.

The final imputation is obtained by adding the predicted residual to the available base measurement:

$$\hat{\mathbf{e}}_{(i,j)} = \mathbf{e}_{(i,j)}^{\text{base}} + \Delta\hat{\mathbf{e}}_{(i,j)}, \tag{48}$$

where $\mathbf{e}(i,j)^{\text{base}}$ denotes the original measurement or the available value before missingness. This residual-based design preserves the physical interpretability of original measurements, mitigates overfitting risks under high missing rates, and enhances prediction robustness.

3.5. Spatiotemporal complexity analysis

In terms of overall time complexity, the proposed model is composed of multiple modules, each introducing different computational costs for feature modeling, spatial relation modeling, and temporal dependency modeling.

First, the missing-aware module mainly consists of linear mappings and gated fusion for node, edge, and mask features, with complexity approximately $O(N_v d_h^2 + N_e d_h^2)$, which is manageable in small- to medium-scale power grids. Next, the mask-enhanced dual-channel attention module computes both main-path and mask-path attention matrices, leading to a complexity of $O(N_e^2 d_h)$. This becomes the primary bottleneck in the spatial dimension, as the quadratic complexity in N_e results in high computational cost when the number of edges is large. For the spatial structure modeling module, the multi-scale heat kernel diffusion GCN branch has a complexity of $O(MKN_e d_h)$, while the EGAT branch costs $O(N_e d_h)$. Together with the gated fusion and node-to-edge restoration operations $O(N_v d_h^2 + N_e d_h^2)$, the overall cost can also be significant in large-scale edge-dense networks. Moving to the temporal dimension, the short-term dynamic modeling part integrates multi-scale dilated convolution, neural ODEs, and forward–backward attention. The convolutional part requires $O(BCTk)$, neural ODE continuous-time solving costs $O(BCTS)$, and attention fusion adds $O(BT^2 d_h + BTd_h^2)$. When implemented as multi-head attention with H heads and per-head dimension $d_k = d_h/H$ (under a fixed d_h), the attention-score computation is $H \cdot O(BT^2 d_k) = O(BT^2 d_h)$, and the $Q/K/V$ plus output projections introduce an additional $O(BTd_h^2)$ term, making the computation substantial for long sequences. For long-term dependency modeling, the Informer-based encoder adopts ProbSparse attention with a complexity of $O(BuTd_h)$, along with the feed-forward network cost $O(BTd_h d_{ff})$; similarly, using H heads under fixed d_h does not change the asymptotic term in $BuTd_h$ (since $H \cdot O(BuTd_k) = O(BuTd_h)$), while the linear projections contribute an additional $O(BTd_h^2)$ per layer. After L stacked layers, the total complexity is $O\left(BL(uTd_h + Td_h d_{ff})\right)$ (plus $O(BLTd_h^2)$ for the projection layers). While this still incurs high computational cost for long sequences, it is significantly more efficient than the standard Transformer's $O(BT^2 d_h)$. Finally, the residual imputation predictor (MLP) only requires $O(BN_e d_h^2)$, which is the lightest component in terms of computational overhead.

In summary, the main computational bottlenecks of the model lie in the dual-channel spatial attention ($O(BN_e^2 d_h)$), the long-term dependency modeling in the temporal dimension ($O(BL(uTd_h + Td_h d_{ff}))$, plus projections $O(BLTd_h^2)$), and the neural ODE solver ($O(BCTS)$). The overall complexity can be expressed as:

$$O\left(BN_e^2 d_h\right) + O\left(BMKN_e d_h\right) + O\left(BL\left(uTd_h + Td_h d_{ff}\right)\right) + O\left(BLTd_h^2\right) + O(BCTS). \tag{49}$$

Here $O(BLTd_h^2)$ accounts for the $Q/K/V$ and output projections in multi-head attention. With a fixed model dimension d_h and H heads ($d_k = d_h/H$), the attention-score computation remains $H \cdot O(BT^2 d_k) = O(BT^2 d_h)$ (and $H \cdot O(BuTd_k) = O(BuTd_h)$ for ProbSparse). Thus, H mainly affects constant factors and the projection overhead, while it does not change the asymptotic order of the score terms under fixed d_h.

In scenarios with medium-scale node sets, large numbers of edges, and long temporal sequences, this complexity structure ensures that the model is capable of capturing complex missing patterns while maintaining feasible computational overhead for practical tasks.

4. Experimental results and analysis

4.1. Datasets

This study evaluates the proposed method on two types of datasets: (i) the NREL-118 benchmark test system with 118 nodes [33], and (ii) a real-world provincial power grid dataset.

The modified IEEE-118 test system dataset released by the U.S. National Renewable Energy Laboratory (NREL) is adopted in this study. This dataset is a simulated benchmark built upon the standard IEEE-118 bus model and is generated via physical-model-based power-system simulations. Importantly, it provides a year-long, time-continuous operational trajectory, which naturally covers diverse seasonal load patterns and different time-of-day operating stages (e.g., off-peak/peak periods). Such full-year coverage enables evaluation of model robustness under realistic non-stationary operating conditions, including long-range variations in demand levels and correlated spatiotemporal dynamics over the grid topology.

The real-world dataset is collected from the operational records of a provincial-level power grid, including 4450 transmission branches, 1613 generators, and 1200 nodes. Data were continuously recorded from December to May (6 months) with a temporal resolution of 5 min, yielding around-the-clock operational trajectories (288 samples per day, approximately 51,840 time steps in total). Notably, this six-month span naturally covers distinct seasonal demand regimes (winter-to-spring transitions) as well as diverse time-of-day operating stages (e.g., off-peak/peak periods), thereby providing non-stationary operating conditions for evaluating model robustness and generalization across changing load patterns and grid dynamics. Due to the sensitivity of real-world power grid information, this dataset is not publicly available.

4.2. Baseline models

BRITS [34]: A bidirectional RNN-based method for time-series imputation, BRITS learns and updates missing values via backpropagation without relying on specific generative assumptions, enabling it to handle multivariate and nonlinear temporal dynamics.

SAITS [22]: A transformer-based imputation model that employs bidirectional diagonal-masked self-attention. By fusing representations from two masked attention blocks, SAITS explicitly models temporal dependencies and feature correlations, achieving both high accuracy and efficiency.

MTC-GNN [35]: An end-to-end model for traffic data imputation. It integrates spatiotemporal feature fusion, adaptive adjacency learning, and masked GRU units to infer missing values from neighboring sensors, thereby achieving accurate recovery.

CC-GAIN [24]: CC-GAIN is a generative adversarial imputation model designed for missing data in power load time series. By introducing sample clustering and pattern classification mechanisms into the traditional GAIN framework, it enables more accurate recovery of power consumption sequences with diverse behavioral patterns under high missing rates.

G-LSTM [36]: G-LSTM is a general spatio-temporal forecasting framework that integrates graph neural networks with long short-term memory networks. By constructing a spatio-temporal graph over the sensor network, it jointly models spatial correlations and temporal dependencies, effectively capturing nonlinear dynamic patterns for modeling and predicting complex time series data.

4.3. Missing data construction

To simulate missing patterns in power system measurements, we construct two types of missingness:

- Random Missing. For edge features (P_{linef}, P_{linet}, Q_{linef}, Q_{linet}), values are randomly masked with missing ratios ranging from 10% to 90% (step size: 10%), simulating unstructured scenarios such as packet loss, transient measurement errors, and momentary sensor faults.
- Continuous Missing. Consecutive missing durations are designed according to the temporal resolution of each dataset to ensure comparable missing proportions across datasets. Specifically, the NREL-118 dataset is sampled hourly, so we mask consecutive segments of 6 h, 12 h, and 18 h (i.e., 6/12/18 time steps) to simulate long-term sensor or communication failures.

In contrast, the real-world dataset is recorded every 5 min; therefore, we use shorter consecutive missing durations of 30 min, 60 min, and 90 min, which correspond to 6/12/18 time steps as well, so that the missing proportions are comparable across datasets. Moreover, to emulate the mixed missingness commonly observed in practical monitoring systems, we further impose an additional 10% random missing ratio over the entire sequence on top of the continuous-missing setting, resulting in hybrid patterns with both random dropouts and sustained outages.

4.4. Hyperparameter settings

To ensure a fair comparison, all baseline models (BRITS, SAITS, MTC-GNN, CC-GAIN, and G-LSTM) are re-trained from scratch on both datasets under the same experimental protocol, including identical data splits (train/validation/test), missingness construction strategies (random and continuous), and evaluation metrics. For each method, we follow the recommended configurations reported in the original papers or public implementations, and keep the network architectures consistent with the official settings (i.e., no structural modifications), while only tuning a small set of training-sensitive hyperparameters. Specifically, the learning rate is selected from $1\times10^{-4}, 3\times10^{-4}, 1\times10^{-3}, 3\times10^{-3}$, weight decay is selected from $0, 1\times10^{-5}, 1\times10^{-4}, 1\times10^{-3}$ (including 0 to cover default settings without weight decay), and the hidden dimension is selected (when applicable) from $64, 128$. All models are trained for up to 300 epochs, and model selection is conducted on the validation set with early stopping enabled to avoid underfitting or overfitting. The final test results are reported using the checkpoint with the best validation performance, thereby reducing systematic underestimation caused by mismatched learning rates, insufficient regularization, or inappropriate model capacity, and ensuring that performance differences mainly reflect modeling capability rather than training-setting discrepancies.

4.5. Evaluation metrics and loss functions

All input and output data were normalized before training to ensure scale consistency. Imputation accuracy was evaluated using the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2}, \quad \mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{y}_i - y_i\right|, \tag{50}$$

where y_i and $\hat{y}_i$ denote the ground-truth and imputed values, respectively. The metrics are computed over all line power features (active/reactive components P, Q) after denormalization, and thus retain the physical units of MVA.

During training, the model was optimized using an MAE-based loss with an additional smoothness regularization to suppress residual fluctuations:

$$\mathcal{L} = \mathcal{L}_{\mathrm{MAE}} + \lambda\mathcal{L}_{\mathrm{smooth}}, \quad \mathcal{L}_{\mathrm{smooth}} = \frac{1}{N}\|\hat{\mathbf{Y}} - \mathbf{Y}\|_2^2, \tag{51}$$

where $\mathbf{Y} = [y_1, \ldots, y_N]^\top$, $\hat{\mathbf{Y}} = [\hat{y}_1, \ldots, \hat{y}_N]^\top$, and $\lambda = 0.01$ controls the contribution of the smoothness term.

4.6. Comparative experimental analysis

To comprehensively validate the effectiveness of the proposed STGNI model, we conducted experiments on both the NREL 118-bus system and a real-world dataset under two types of missingness patterns: random missing and continuous missing. We compared STGNI with mainstream imputation methods including BRITS, SAITS, MTC-GNN, CC-GAIN and G-LSTM, using MAE and RMSE as evaluation metrics. The results are reported in the following tables.

4.6.1. Random missing experimental

Under random missing patterns, both Figs. 3 and 4, and the extracted results in Table 2 reveal distinct behaviors on NREL-118 and the real-world dataset. On NREL-118, most methods maintain relatively high correlation as the missing rate increases, reflecting the strong regularity and spatial consistency of this benchmark system; however, MTC-GNN degrades the most, with MAE increasing from 0.0540 at 30% missing to 0.0889 at 90% missing and R^2 dropping from 0.9394 to 0.8694, indicating that neighborhood message passing becomes unreliable when a large portion of node features is missing. In contrast, STGNI remains consistently stable, achieving the lowest MAE and RMSE and the highest PCC and R^2 across all missing rates (e.g., R^2 remains above 0.997 even at 90% missing), which suggests that topology-consistent spatial regularization combined with robust temporal modeling effectively mitigates error accumulation. On the real-world dataset, performance differences across methods become more pronounced as the missing rate increases, mainly due to inevitable measurement noise and stronger temporal heterogeneity and non-stationary dynamics. BRITS and SAITS, which rely primarily on temporal continuity, exhibit substantially larger errors and lower correlations, particularly at high missing rates, as random missing disrupts short-term temporal context and leads to cumulative prediction drift. MTC-GNN performs comparatively better than purely temporal baselines by exploiting local spatial correlations, but its MAE, PCC, and R^2 still deteriorate noticeably at high missing rates, indicating

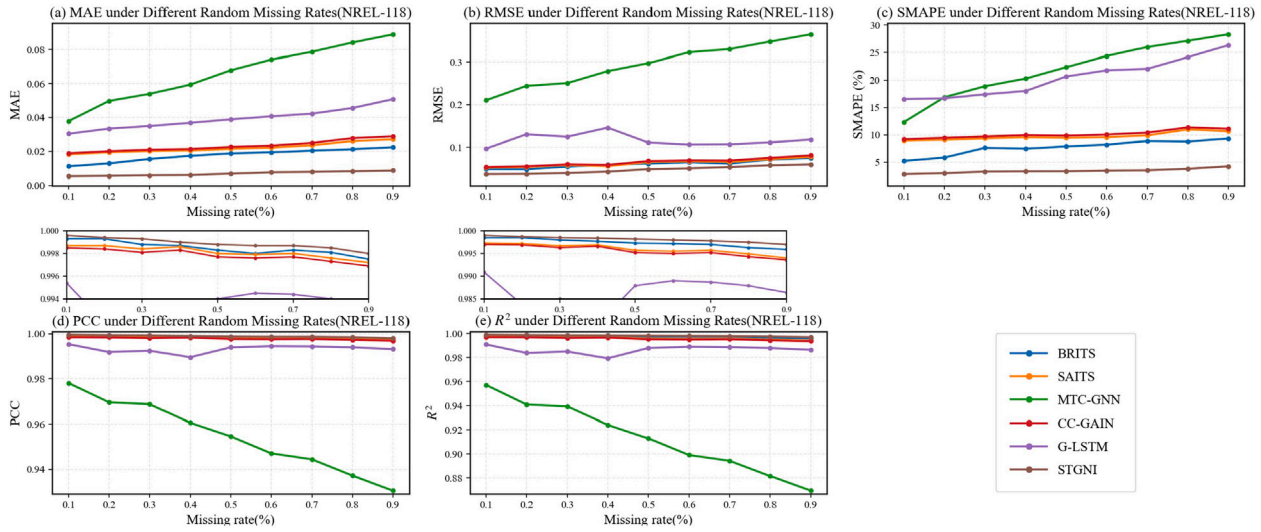


Fig. 3. Imputation performance comparison under different random missing rates on NREL-118 dataset.

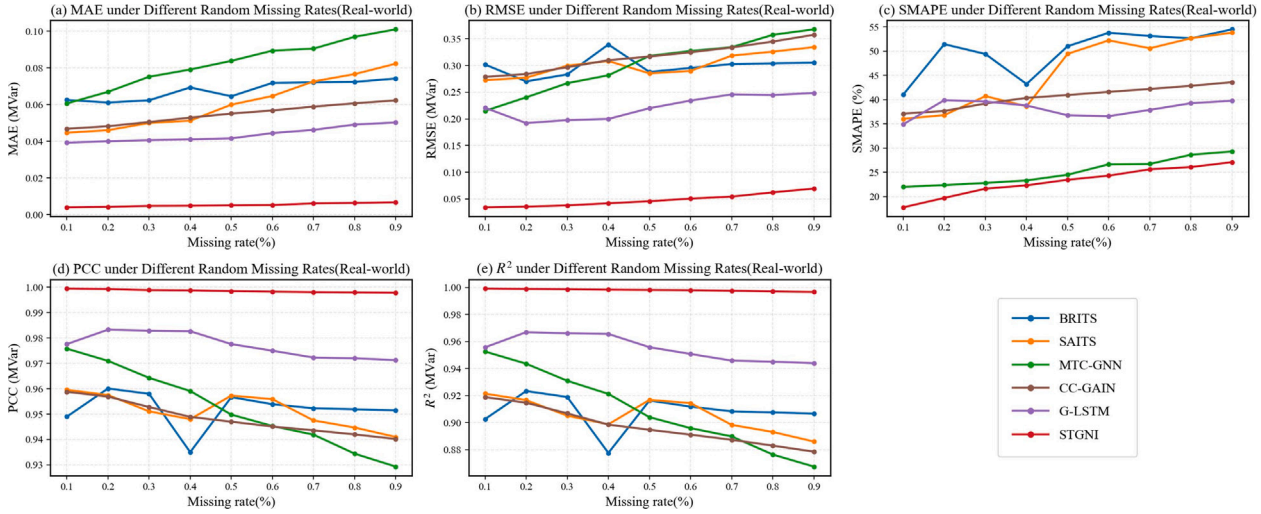


Fig. 4. Imputation performance comparison under different random missing rates on Real-World dataset.

limited robustness under severe observation sparsity. Overall, the consistently superior performance of STGNI on both datasets demonstrates that jointly modeling topology-aware spatial constraints and short–long-term temporal dependencies is critical for robust imputation under high random missing rates, especially in real-world scenarios with complex and noisy dynamics.

4.6.2. *Continuous missing experimental*

As shown in Table 3, under the continuous-missing setting on both datasets, most methods degrade as the missing duration increases: MAE and RMSE rise while PCC and R^2 decline, indicating that long missing segments disrupt temporal context and lead to error accumulation. On NREL-118, for example, BRITS' MAE increases from 0.0195 (6 h) to 0.0299 (18 h), while its R^2 decreases from 0.9969 to 0.9962.

On NREL-118, purely temporal baselines (BRITS and SAITS) deteriorate steadily as the gap length increases; meanwhile, MTC-GNN, which relies on neighborhood message passing, is affected the most when local neighborhood information is unavailable for a prolonged period (MAE increases from 0.0620 to 0.0712, while PCC decreases from 0.9668 to 0.9618). CC-GAIN and G-LSTM perform at an intermediate level: CC-GAIN's MAE is 0.0217, 0.0240, and 0.0286 and its PCC is 0.9981, 0.9981, and 0.9972 under 6 h, 12 h, and 18 h, showing a stable degradation with longer durations; G-LSTM's MAE is 0.0346, 0.0356, and 0.0394 and its PCC is 0.9923, 0.9947, and 0.9945, which is more stable than MTC-GNN but with higher errors. In contrast, STGNI remains the best and relatively stable across durations (MAE is 0.0077, 0.0082, and 0.0116 for 6 h, 12 h, and 18 h, and R^2 drops from 0.9984 to 0.9909), indicating that it can effectively suppress unreliable signals and stabilize information propagation under long missing segments. Moreover, on the spatial side, the dual-channel structure learning employs two complementary mechanisms, diffusion-based propagation and edge-guided attention, to provide topology-consistent regularization; on the temporal side, the short–long-term encoder jointly captures within-window dynamics and long-range dependencies, which is particularly important when long gaps break short-range temporal context.

On the real-world dataset, STGNI achieves the best overall performance because it jointly leverages topology-aware spatial constraints and temporal dependency modeling. This design suppresses unreliable signals and stabilizes information propagation under long continuous-missing segments, leading to smaller errors and stronger trend consistency (e.g., under 30, 60, and 90 min missing, MAE is 0.0077, 0.0085, and 0.0090; PCC is 0.9974, 0.9976, and 0.9953; and R^2 is 0.9947,

Table 2
Performance comparison under random missing patterns at 30%, 60%, and 90% missing rates.

Dataset	Missing	Metric	BRITS	SAITS	MTC-GNN	CC-GAIN	G-LSTM	STGNI
NREL-118	Random (30%)	MAE	0.0155	0.0202	0.0540	0.0210	0.0350	**0.0059**
		RMSE	0.0550	0.0586	0.2508	0.0601	0.1250	**0.0400**
		SMAPE(%)	7.62	9.35	18.87	9.70	17.40	**3.28**
		PCC	0.9988	0.9984	0.9688	0.9981	0.9925	**0.9993**
		R^2	0.9980	0.9967	0.9394	0.9963	0.9850	**0.9985**
	Random (60%)	MAE	0.0194	0.0222	0.0741	0.0234	0.0407	**0.0077**
		RMSE	0.0650	0.0674	0.3229	0.0693	0.1065	**0.0508**
		SMAPE(%)	8.20	9.58	24.37	10.05	21.77	**3.47**
		PCC	0.9980	0.9979	0.9471	0.9976	0.9945	**0.9987**
		R^2	0.9972	0.9955	0.8991	0.9950	0.9890	**0.9980**
	Random (90%)	MAE	0.0224	0.0271	0.0889	0.0289	0.0508	**0.0087**
		RMSE	0.0749	0.0780	0.3649	0.0812	0.1183	**0.0600**
		SMAPE(%)	9.32	10.66	28.33	11.10	26.32	**4.25**
		PCC	0.9975	0.9972	0.9305	0.9969	0.9932	**0.9980**
		R^2	0.9959	0.9940	0.8694	0.9936	0.9864	**0.9970**
Real-world	Random (30%)	MAE	0.0623	0.0500	0.0752	0.0505	0.0406	**0.0048**
		RMSE	0.2833	0.2997	0.2668	0.2968	0.1976	**0.0378**
		SMAPE(%)	49.41	40.71	22.78	39.20	39.60	**21.60**
		PCC	0.9580	0.9511	0.9643	0.9528	0.9829	**0.9989**
		R^2	0.9190	0.9052	0.9310	0.9069	0.9661	**0.9986**
	Random (60%)	MAE	0.0718	0.0647	0.0894	0.0568	0.0445	**0.0052**
		RMSE	0.2955	0.2899	0.3272	0.3247	0.2343	**0.0506**
		SMAPE(%)	53.81	52.24	26.64	41.60	36.58	**24.29**
		PCC	0.9539	0.9559	0.9453	0.9451	0.9750	**0.9983**
		R^2	0.9119	0.9145	0.8960	0.8912	0.9508	**0.9979**
	Random (90%)	MAE	0.0742	0.0823	0.1010	0.0623	0.0502	**0.0068**
		RMSE	0.3053	0.3344	0.3677	0.3576	0.2484	**0.0694**
		SMAPE(%)	54.51	53.83	29.29	43.60	39.76	**27.08**
		PCC	0.9515	0.9410	0.9293	0.9402	0.9713	**0.9979**
		R^2	0.9068	0.8861	0.8676	0.8786	0.9440	**0.9966**

Table 3
Performance under continuous missing durations.

Dataset	Duration	Metric	Model					
			BRITS	SAITS	MTC-GNN	CC-GAIN	G-LSTM	STGNI
NREL-118	6 h	MAE	0.0195	0.0210	0.0620	0.0217	0.0346	**0.0077**
		RMSE	0.0562	0.0605	0.2583	0.0620	0.1265	**0.0417**
		SMAPE(%)	8.13	8.80	22.29	9.10	17.03	**4.02**
		PCC	0.9984	0.9983	0.9668	0.9981	0.9923	**0.9992**
		R^2	0.9969	0.9965	0.9354	0.9961	0.9847	**0.9984**
	12 h	MAE	0.0202	0.0232	0.0668	0.0240	0.0356	**0.0082**
		RMSE	0.0579	0.0606	0.2683	0.0621	0.1028	**0.0478**
		SMAPE(%)	8.43	9.85	24.78	10.20	19.12	**3.87**
		PCC	0.9983	0.9984	0.9627	0.9981	0.9947	**0.9990**
		R^2	0.9967	0.9963	0.9275	0.9960	0.9895	**0.9978**
	18 h	MAE	0.0299	0.0270	0.0712	0.0286	0.0394	**0.0116**
		RMSE	0.0623	0.0719	0.2714	0.0735	0.1054	**0.0972**
		SMAPE(%)	9.96	10.49	26.63	10.95	21.06	**3.93**
		PCC	0.9981	0.9976	0.9618	0.9972	0.9945	**0.9958**
		R^2	0.9962	0.9949	0.9258	0.9943	0.9890	**0.9909**
Real-world	30 min	MAE	0.1027	0.0807	0.0152	0.0760	0.0689	**0.0077**
		RMSE	0.4103	0.3762	0.1046	0.3600	0.3021	**0.0745**
		SMAPE(%)	50.40	45.83	28.52	44.80	46.04	**24.60**
		PCC	0.9062	0.9222	0.9948	0.9450	0.9561	**0.9974**
		$\mathbf{R^2}$	0.8242	0.8519	0.9897	0.9050	0.9146	**0.9947**
	60 min	MAE	0.0997	0.0727	0.0141	0.0685	0.0559	**0.0085**
		RMSE	0.3972	0.3568	0.1012	0.3450	0.2606	**0.0700**
		SMAPE(%)	49.94	43.28	31.70	41.80	42.61	**28.91**
		PCC	0.9124	0.9307	0.9951	0.9520	0.9674	**0.9976**
		$\mathbf{R^2}$	0.8350	0.8669	0.9902	0.9100	0.9358	**0.9952**
	90 min	MAE	0.1486	0.0702	0.0142	0.0660	0.0738	**0.0090**
		RMSE	0.4634	0.3404	0.1067	0.3320	0.2603	**0.0988**
		SMAPE(%)	65.38	42.27	32.23	40.90	52.35	**24.46**
		PCC	0.8842	0.9373	0.9945	0.9550	0.9682	**0.9953**
		$\mathbf{R^2}$	0.7816	0.8793	0.9890	0.9200	0.9373	**0.9906**

0.9952, and 0.9906). MTC-GNN also performs competitively because neighborhood message passing can effectively exploit local spatial correlations, and thus it achieves relatively low RMSE across durations (0.1046, 0.1012, and 0.1067 for 30, 60, and 90 min). However, when the missing segment becomes long, relying mainly on local propagation is more prone to cumulative drift, so its MAE and correlation metrics remain inferior to STGNI (e.g., at 60 min, MAE is 0.0141, PCC is 0.9951, and R^2 is 0.9902). In contrast, BRITS and SAITS are purely temporal imputation baselines that depend more on short-term temporal context; continuous missing breaks temporal continuity, induces error accumulation, and lacks topology redundancy for regularization, resulting in noticeably worse performance (e.g., at 90 min, BRITS has RMSE of 0.4634 and PCC of 0.8842, while SAITS has RMSE of 0.3404 and PCC of 0.9373).

4.7. Ablation study

To evaluate the practical value of the key designs in STGNI, we conducted seven types of ablation experiments by systematically removing core components and analyzing their impact on overall performance: Ablation-1 (Bidirectional Neural ODE Temporal Encoder): Remove the bidirectional neural ODE encoder, keeping only the long-term encoder. Ablation-2 (Long-term Temporal Encoder): Remove the long-term encoder, retaining only the bidirectional neural ODE encoder. Ablation-3 (Multi-scale Heat Kernel GCN): Remove the multi-scale heat kernel GCN, keeping only the edge-guided graph attention module. Ablation-4 (Edge-guided Graph Attention): Remove the edge-guided attention module, retaining only the multi-scale heat kernel GCN. Ablation-5 (Missing-aware Module): Remove the missing-aware components, while keeping the hybrid temporal encoder and dual-channel spatial modeling. Ablation-6 (Spatial Module Removed): Remove the combined spatial modeling block (i.e., both the Multi-scale Heat Kernel GCN and the Edge-guided Graph Attention modules), while keeping the hybrid temporal encoder and the missing-aware components. Ablation-7 (Temporal Module Removed): Remove the entire temporal modeling block (i.e., both the Bidirectional Neural ODE Temporal Encoder and the Long-term Temporal Encoder), while keeping the dual-channel spatial modeling and the missing-aware components.

4.7.1. Ablation study under random missingness

Combining Figs. 5 and 6, and Table 4, the full STGNI achieves the lowest MAE and RMSE and maintains relatively stable PCC and R^2

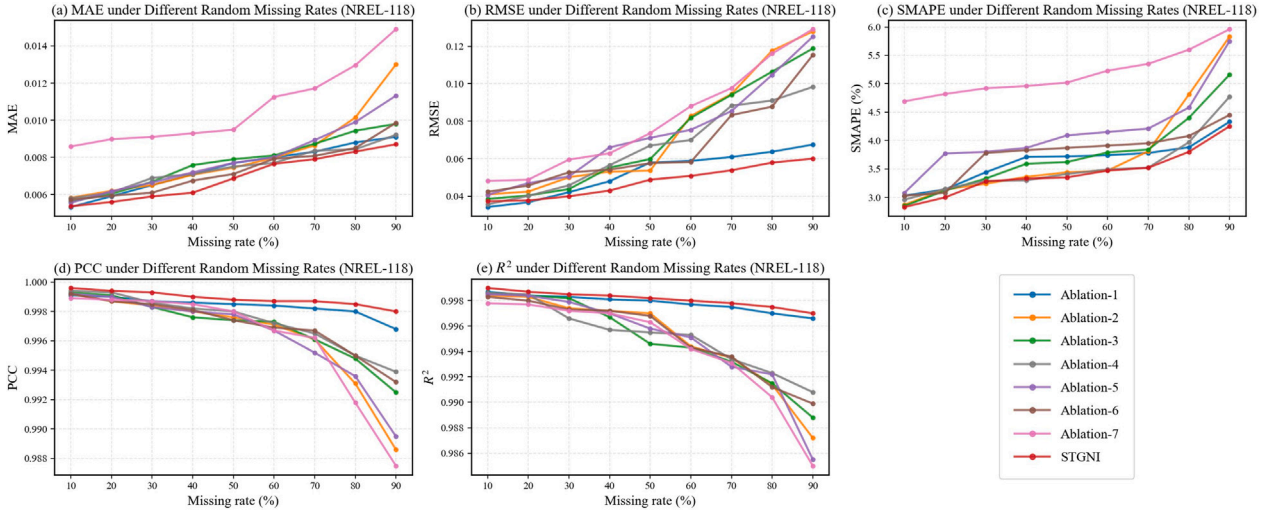


Fig. 5. Imputation performance comparison under different random missing rates on the NREL-118 dataset with Ablation Analysis.

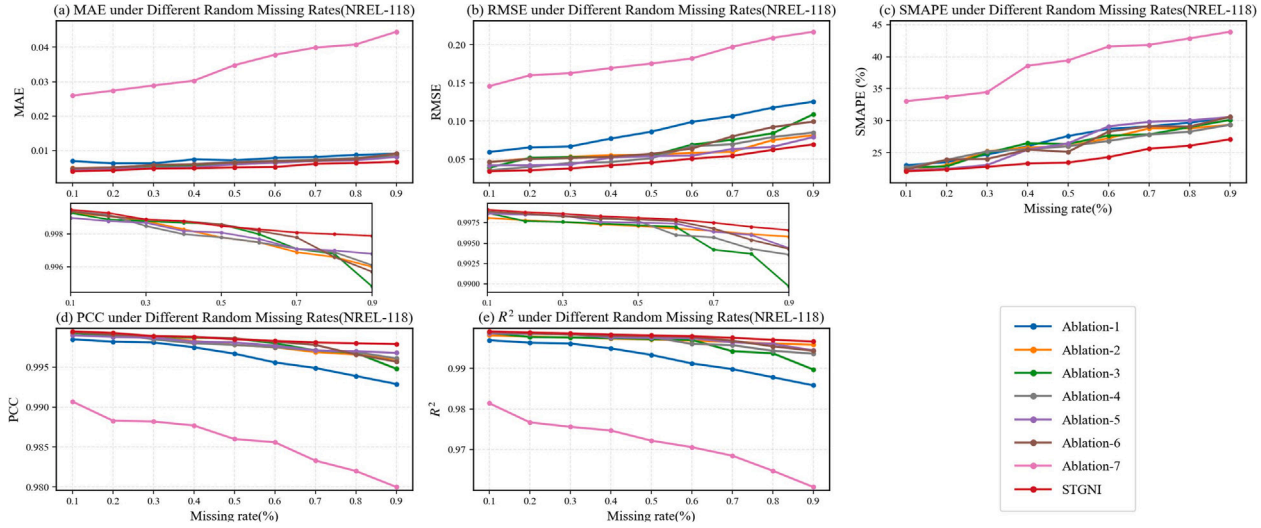


Fig. 6. Imputation performance comparison under different random missing rates on the Real-World dataset with Ablation Analysis.

Table 4
Ablation performance under random missing patterns at 30%, 60%, and 90% missing rates.

Dataset	Missing	Metric	Ablation-1	Ablation-2	Ablation-3	Ablation-4	Ablation-5	Ablation-6	Ablation-7	STGNI
Real-World	Random (30%)	MAE	0.0063	0.0052	0.0051	0.0058	0.0049	0.0058	0.0289	**0.0048**
		RMSE	0.0667	0.0531	0.0529	0.0451	0.0430	0.0515	0.1628	**0.0378**
		SMAPE(%)	24.69	25.21	24.93	25.19	23.05	24.01	34.45	**22.78**
		PCC	0.9981	0.9988	0.9988	0.9985	0.9987	0.9989	0.9882	**0.9989**
		R^2	0.9961	0.9976	0.9976	0.9983	0.9984	0.9983	0.9756	**0.9986**
	Random (60%)	MAE	0.0079	0.0065	0.0066	0.0070	0.0065	0.0069	0.0378	**0.0052**
		RMSE	0.0988	0.0583	0.0690	0.0669	0.0547	0.0639	0.1819	**0.0506**
		SMAPE(%)	28.71	27.25	27.67	26.82	29.10	28.28	41.58	**24.29**
		PCC	0.9956	0.9975	0.9980	0.9975	0.9977	0.9982	0.9856	**0.9983**
		R^2	0.9912	0.9968	0.9970	0.9960	0.9974	0.9977	0.9706	**0.9979**
	Random (90%)	MAE	0.0091	0.0085	0.0082	0.0089	0.0081	0.0091	0.0445	**0.0068**
		RMSE	0.1254	0.0814	0.1087	0.0849	0.0790	0.0993	0.2169	**0.0694**
		SMAPE(%)	30.51	29.34	30.11	29.36	30.55	30.60	43.89	**27.08**
		PCC	0.9929	0.9960	0.9948	0.9961	0.9968	0.9957	0.9800	**0.9979**
		R^2	0.9858	0.9958	0.9897	0.9936	0.9944	0.9943	0.9608	**0.9966**
NREL-118	Random (30%)	MAE	0.0065	0.0065	0.0067	0.0069	0.0067	0.0061	0.0091	**0.0059**
		RMSE	0.0422	0.0501	0.0438	0.0458	0.0505	0.0527	0.0595	**0.0400**
		SMAPE(%)	3.44	3.24	3.33	3.29	3.80	3.78	4.92	**3.28**
		PCC	0.9987	0.9984	0.9983	0.9986	0.9983	0.9985	0.9987	**0.9993**
		R^2	0.9983	0.9974	0.9982	0.9966	0.9979	0.9973	0.9972	**0.9985**
	Random (60%)	MAE	0.0080	0.0080	0.0081	0.0077	0.0080	0.0079	0.0113	**0.0077**
		RMSE	0.0588	0.0828	0.0818	0.0700	0.0754	0.0582	0.0880	**0.0508**
		SMAPE(%)	3.74	3.47	3.79	3.49	4.15	3.91	5.23	**3.47**
		PCC	0.9984	0.9971	0.9973	0.9972	0.9967	0.9969	0.9967	**0.9987**
		R^2	0.9977	0.9944	0.9943	0.9953	0.9951	0.9943	0.9942	**0.9980**
	Random (90%)	MAE	0.0091	0.0130	0.0098	0.0092	0.0113	0.0099	0.0149	**0.0087**
		RMSE	0.0675	0.1279	0.1188	0.0983	0.1251	0.1153	0.1291	**0.0600**
		SMAPE(%)	4.33	5.83	5.16	4.77	5.75	4.45	5.96	**4.25**
		PCC	0.9968	0.9886	0.9925	0.9939	0.9895	0.9932	0.9875	**0.9980**
		R^2	0.9966	0.9872	0.9888	0.9908	0.9855	0.9899	0.9850	**0.9970**

across missing rates on both datasets. Once any key module is removed, both accuracy and correlation degrade systematically, and the degradation becomes more pronounced at 60% and 90% missingness. On NREL-118, at 30%, 60%, and 90% missingness, STGNI attains RMSE values of 0.0400, 0.0508, and 0.0600, and PCC values of 0.9993, 0.9987, and 0.9980. The temporal module is crucial for trend preservation: at 90% missingness, Ablation-1 and Ablation-2 increase RMSE to 0.0675 and 0.1279 and reduce PCC to 0.9968 and 0.9886, indicating that a single temporal pathway cannot simultaneously capture short-term dynamics and long-range dependencies when observations are sparse. Ablation-7 degrades the most, with RMSE of 0.1291 and PCC of 0.9875 at 90% missingness, showing that severe missingness requires explicit temporal consistency to constrain global evolution. Spatial modeling is also indispensable: at 60% missingness, Ablation-3 and Ablation-4 yield RMSE values of 0.0818 and 0.0700, both higher than STGNI's 0.0508, suggesting that retaining only one spatial mechanism is insufficient to jointly characterize multi-scale topological redundancy and local edge correlations. Ablation-6 further increases errors under high missingness, reaching an RMSE of 0.1153 at 90% missingness, which reflects the difficulty of robustly completing information from neighbors without adequate topological constraints. The missingness-aware mechanism determines robustness at high missing rates: on NREL-118 at 90% missingness, Ablation-5 achieves an RMSE of 0.1251 and a PCC of 0.9895, which is clearly worse than STGNI with an RMSE of 0.0600 and a PCC of 0.9980, implying that without distinguishing reliable observations from uncertain inputs the model is more vulnerable to mask-sampling variance and noise. This pattern is even more evident on the real-grid data: at 30%, 60%, and 90% missingness, STGNI attains RMSE values of 0.0378, 0.0506, and 0.0694, and PCC values of 0.9989, 0.9983, and 0.9979, whereas Ablation-7 reaches RMSE values of 0.1628, 0.1819, and 0.2169 and PCC drops to 0.9882, 0.9856, and 0.9800. These results indicate that under stronger non-stationarity and distribution shift, spatio-temporal joint constraints are more critical for preserving trends. The smoother curves and slower correlation decay of STGNI further reflect the synergistic gains from reliability modeling, dual-branch spatial structure modeling, and hybrid temporal encoding.

4.7.2. Ablation study under continuous missingness

Table 5 reports the ablation results on NREL-118 and the real-world dataset under different continuous missing durations. Overall, as the missing duration increases, most ablated variants exhibit larger MAE and RMSE, together with lower PCC and R^2. In contrast, the full STGNI maintains the most robust errors and correlations across both datasets and all durations.

On NREL-118, the impact of removing temporal components is most evident. Ablation-2, which removes the long-term encoder, increases RMSE from 0.0498 at 6 h to 0.1031 at 12 h, whereas STGNI achieves an RMSE of 0.0478 at 12 h, indicating that without long-range trend constraints the model is more prone to drift within long missing segments. Spatial modeling is also crucial. Ablation-4, which removes the edge-guided attention, reaches an RMSE of 0.0813 at 6 h, substantially higher than the RMSE of 0.0417 achieved by STGNI, suggesting that the lack of edge-level local interactions weakens neighborhood compensation. Moreover, Ablation-6 yields RMSE values of 0.0482, 0.0749, and 0.1199 at 6 h, 12 h, and 18 h, respectively, all higher than those of STGNI, highlighting the importance of enforcing topology-consistent constraints.

On the real-world dataset, the degradation becomes more severe, which more clearly demonstrates the indispensability of the modules. Ablation-7 leads to MAE values of 0.0491, 0.0496, and 0.0472 and RMSE values of 0.2208, 0.2068, and 0.2054 at 30 min, 60 min, and 90 min, respectively. Meanwhile, its PCC decreases to 0.9771, 0.9799, and 0.9802, which is markedly worse than STGNI under the corresponding durations, where MAE is 0.0077, 0.0085, and 0.0090, RMSE is 0.0745, 0.0700, and 0.0988, and PCC is at least 0.9953. These results indicate that without temporal evolution constraints, the model becomes highly unstable

Table 5
Ablation performance under different continuous missing durations on the NREL-118 and real-world datasets.

Dataset	Duration	Metric	Model							
			Ablation-1	Ablation-2	Ablation-3	Ablation-4	Ablation-5	Ablation-6	Ablation-7	STGNI
NREL-118	6 h	MAE	0.0097	0.0088	0.0092	0.0109	0.0103	0.0085	0.0150	**0.0077**
		RMSE	0.0523	0.0498	0.0606	0.0813	0.0804	0.0482	0.0651	**0.0417**
		SMAPE(%)	4.27	3.91	3.89	4.00	3.95	3.83	7.65	**4.02**
		PCC	0.9988	0.9989	0.9984	0.9971	0.9971	0.9989	0.9980	**0.9992**
		R^2	0.9974	0.9976	0.9965	0.9936	0.9938	0.9978	0.9960	**0.9984**
	12 h	MAE	0.0118	0.0127	0.0120	0.0115	0.0123	0.0105	0.0164	**0.0082**
		RMSE	0.0587	0.1031	0.1065	0.0925	0.1016	0.0749	0.0612	**0.0478**
		SMAPE(%)	6.77	4.37	4.18	4.14	4.97	4.48	9.26	**3.87**
		PCC	0.9985	0.9952	0.9949	0.9962	0.9952	0.9975	0.9982	**0.9990**
		R^2	0.9966	0.9897	0.9891	0.9917	0.9897	0.9946	0.9964	**0.9978**
	18 h	MAE	0.0156	0.0151	0.0158	0.0138	0.0163	0.0152	0.0173	**0.0116**
		RMSE	0.1291	0.1105	0.1231	0.1053	0.1309	0.1199	0.0975	**0.0972**
		SMAPE(%)	5.97	6.46	7.04	6.93	5.67	6.76	9.53	**3.93**
		PCC	0.9921	0.9943	0.9928	0.9948	0.9918	0.9932	0.9980	**0.9958**
		R^2	0.9833	0.9878	0.9848	0.9889	0.9828	0.9856	0.9958	**0.9909**
Real-World	30 min	MAE	0.0100	0.0089	0.0076	0.0088	0.0080	0.0083	0.0491	**0.0077**
		RMSE	0.1080	0.0782	0.0721	0.0898	0.0798	0.0809	0.2208	**0.0745**
		SMAPE(%)	25.02	25.05	24.49	24.80	24.68	24.76	41.58	**24.60**
		PCC	0.9947	0.9971	0.9975	0.9962	0.9970	0.9969	0.9771	**0.9974**
		R^2	0.9893	0.9942	0.9951	0.9924	0.9940	0.9938	0.9549	**0.9947**
	60 min	MAE	0.0103	0.0100	0.0082	0.0092	0.0087	0.0091	0.0496	**0.0085**
		RMSE	0.1125	0.0933	0.0827	0.0965	0.0943	0.0886	0.2068	**0.0700**
		SMAPE(%)	24.89	25.33	24.26	24.84	24.41	24.51	41.64	**28.91**
		PCC	0.9942	0.9957	0.9967	0.9956	0.9958	0.9963	0.9799	**0.9976**
		R^2	0.9884	0.9911	0.9934	0.9911	0.9915	0.9946	0.9601	**0.9952**
	90 min	MAE	0.0118	0.0120	0.0103	0.0102	0.0098	0.0109	0.0472	**0.0090**
		RMSE	0.1093	0.1378	0.1104	0.1029	0.0996	0.1081	0.2054	**0.0988**
		SMAPE(%)	25.22	25.05	24.56	24.73	24.58	24.83	40.44	**24.46**
		PCC	0.9946	0.9909	0.9941	0.9948	0.9944	0.9943	0.9802	**0.9953**
		R^2	0.9889	0.9818	0.9881	0.9896	0.9898	0.9886	0.9607	**0.9906**

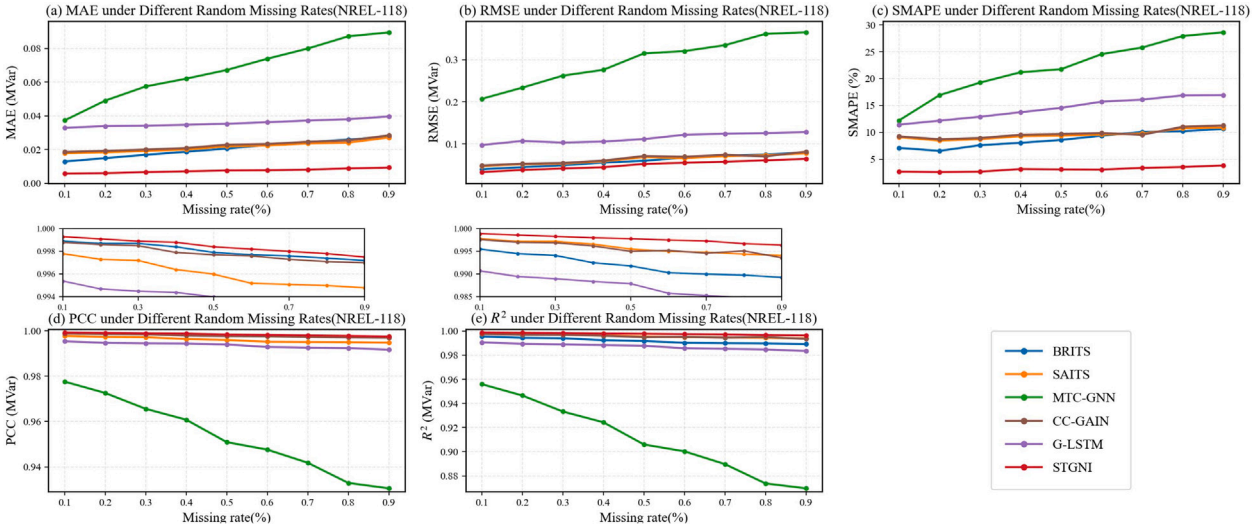


Fig. 7. Imputation performance comparison under the small-noise condition on the NREL-118 dataset.

under noise and non-stationarity. In summary, under continuous missingness, temporal constraints mitigate long-gap drift, spatial constraints enforce topology-consistent recovery, and missingness-aware modeling filters unreliable inputs. Their synergy enables STGNI to maintain the lowest errors and the most stable correlations under long missing durations, and removing any of these components leads to performance degradation.

4.8. Missingness-and-noise coupled setting

To better reflect realistic power-system measurement conditions, we inject measurement noise on top of the constructed missingness patterns, jointly simulating data loss caused by communication dropouts/sensor failures and measurement errors induced by instrument inaccuracy and sampling jitter. Following Zhang et al. [37], we add Gaussian white noise

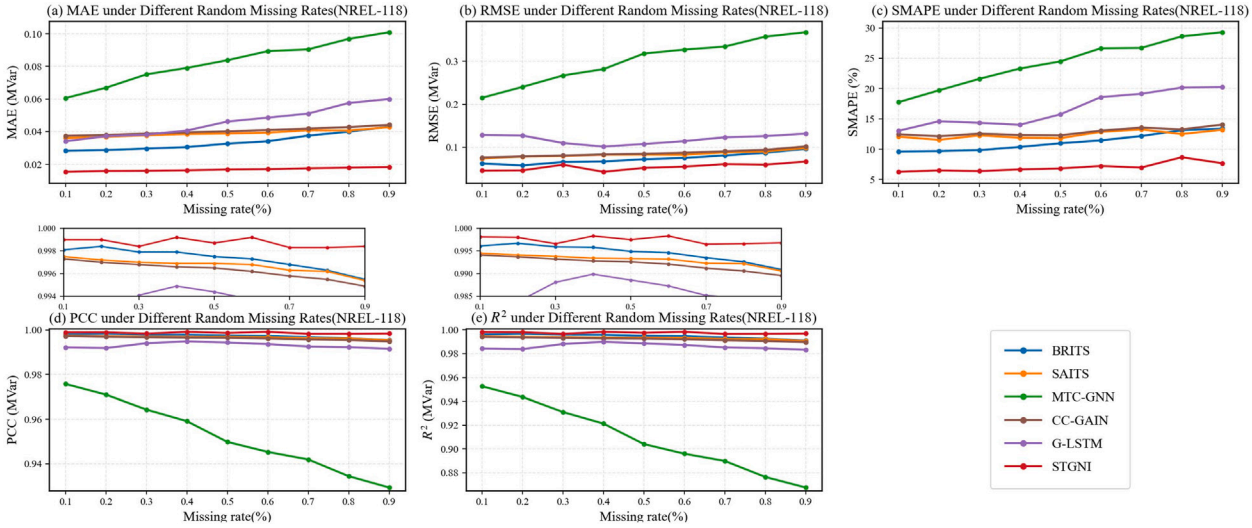


Fig. 8. Imputation performance comparison under the large-noise condition on the NREL-118 dataset.

Table 6
Performance under random missingness with two noise levels at missing rates of 30%, 60%, and 90%.

Noise	Dataset	Missing	Metric	BRITS	SAITS	MTC-GNN	CC-GAIN	G-LSTM	STGNI
Small	NREL-118	Random (30%)	MAE	0.0169	0.0191	0.0574	0.0200	0.0341	**0.0066**
			RMSE	0.0495	0.0534	0.2622	0.0549	0.1033	**0.0418**
			SMAPE(%)	7.61	8.73	19.26	8.95	12.91	**2.68**
			PCC	0.9987	0.9972	0.9656	0.9985	0.9945	**0.9989**
			R^2	0.9941	0.9972	0.9333	0.9969	0.9890	**0.9983**
		Random (60%)	MAE	0.0227	0.0224	0.0738	0.0233	0.0362	**0.0077**
			RMSE	0.0680	0.0663	0.3204	0.0698	0.1220	**0.0554**
			SMAPE(%)	9.38	9.60	24.56	9.88	15.71	**3.07**
			PCC	0.9977	0.9952	0.9476	0.9976	0.9929	**0.9982**
			R^2	0.9903	0.9950	0.9003	0.9952	0.9858	**0.9975**
		Random (90%)	MAE	0.0276	0.0271	0.0895	0.0284	0.0396	**0.0093**
			RMSE	0.0800	0.0778	0.3650	0.0819	0.1282	**0.0646**
			SMAPE(%)	10.65	10.88	28.63	11.28	16.92	**3.84**
			PCC	0.9972	0.9948	0.9305	0.9970	0.9917	**0.9975**
			R^2	0.9893	0.9941	0.8696	0.9936	0.9835	**0.9964**
Large	NREL-118	Random (30%)	MAE	0.0297	0.0378	0.0752	0.0388	0.0382	**0.0161**
			RMSE	0.0653	0.0794	0.2668	0.0805	0.1099	**0.0592**
			SMAPE(%)	9.85	12.27	21.60	12.55	14.34	**6.39**
			PCC	0.9979	0.9970	0.9643	0.9968	0.9941	**0.9984**
			R^2	0.9959	0.9938	0.9310	0.9932	0.9881	**0.9966**
		Random (60%)	MAE	0.0342	0.0394	0.0894	0.0410	0.0486	**0.0171**
			RMSE	0.0750	0.0823	0.3272	0.0870	0.1142	**0.0547**
			SMAPE(%)	11.46	12.85	26.64	13.05	18.58	**7.24**
			PCC	0.9973	0.9968	0.9453	0.9962	0.9937	**0.9992**
			R^2	0.9946	0.9932	0.8960	0.9921	0.9873	**0.9983**
		Random (90%)	MAE	0.0431	0.0428	0.1010	0.0442	0.0600	**0.0184**
			RMSE	0.0961	0.0981	0.3677	0.1018	0.1313	**0.0664**
			SMAPE(%)	13.38	13.19	29.29	14.05	20.25	**7.69**
			PCC	0.9955	0.9954	0.9293	0.9949	0.9916	**0.9984**
			R^2	0.9909	0.9906	0.8676	0.9896	0.9832	**0.9968**

to both active- and reactive-power measurements. In the small-noise condition, the active-power noise has mean/variance (0.19%, 0.04%), while the reactive-power noise has mean/variance (-0.10%, 0.08%). To stress-test robustness under extremely low signal-to-noise ratios, we construct a large-noise condition by scaling both the noise mean and variance by 100×. Under this coupled "missingness–noise" setting, we adopt random missingness and sweep the missing rate from 10% to 90% with a step size of 10%, evaluating all methods under both noise levels. This protocol enables a systematic assessment of error growth and trend-consistency degradation as observations become sparser and noise becomes stronger, thereby testing robustness and generalization under practically realistic disturbances.

Combining Figs. 7 and 8 and Table 6, under the coupled random-missingness and measurement-noise setting on NREL-118, increasing the missing rate from 10% to 90% generally leads to larger MAE, RMSE, and SMAPE, while PCC and R^2 tend to decrease. Moreover, moving from the small-noise case to the large-noise case results in additional degradation for most methods, indicating that sparse observations and stronger measurement errors jointly exacerbate reconstruction difficulty. This error-amplification issue is particularly evident for

message-passing-heavy MTC-GNN: its RMSE stays at a much higher level across both noise conditions, reaching 0.2622 under small noise and 0.2668 under large noise at 30% missingness, and further increasing to 0.3650 under small noise and 0.3677 under large noise at 90% missingness, because neighborhood aggregation can propagate and accumulate corrupted or uncertain signals along the topology when measurement reliability is not explicitly controlled. Purely temporal imputers such as BRITS and SAITS also exhibit clear error accumulation as missingness grows; for example, BRITS increases its RMSE from 0.0495 to 0.0800 under small noise and from 0.0653 to 0.0961 under large noise when moving from 30% to 90% missingness, since long gaps break local temporal context and noise-induced deviations drift over time without topology-based correction. In contrast, STGNI maintains the smallest errors and the most stable correlations across both noise levels; for instance, at 90% missingness it attains an RMSE of 0.0646 under small noise and 0.0664 under large noise, which is substantially lower than the corresponding results of BRITS, SAITS, and G-LSTM. This robustness is mainly attributed to three coupled designs: missingness-aware reliability modeling down-weights uncertain or noisy measurements before and during propagation, topology-constrained spatial completion prevents blindly spreading corrupted information along transmission lines, and temporal consistency learning mitigates drift under long missing segments, thereby suppressing error amplification while preserving trend consistency with consistently high PCC and R^2.

4.9. Optimizer comparison

To evaluate the impact of optimizer choice on training, we compare Adam, RMSProp, SGD, and the adopted AdamW under continuous missingness on both the NREL-118 test system and a real provincial power grid dataset. The results are reported in Table 7, where STGNI denotes the default setting using AdamW. Overall, AdamW delivers the best accuracy and the strongest stability across durations, whereas switching to Adam, RMSProp, or SGD leads to consistent performance degradation. We note that a few metrics may exhibit minor non-monotonic fluctuations across durations due to randomness in mask sampling and training dynamics; nevertheless, the relative ranking among optimizers remains stable.

Table 7
Performance of STGNI with different optimizers under continuous missingness on the NREL-118 and real-world datasets.

Dataset	Duration	Metric	Optimizer			
			Adam	RMSProp	SGD	AdamW (our model)
NREL-118	6 h	MAE	0.0137	0.0166	0.0839	0.0077
		RMSE	0.1230	0.1425	0.2963	0.0417
		SMAPE(%)	4.31	5.05	18.23	4.02
		PCC	0.9931	0.9907	0.9627	0.9992
		R^2	0.9854	0.9804	0.9160	0.9984
	12 h	MAE	0.0143	0.0180	0.0943	0.0082
		RMSE	0.1311	0.1435	0.3207	0.0478
		SMAPE(%)	4.24	6.24	19.96	3.87
		PCC	0.9921	0.9903	0.9563	0.9990
		R^2	0.9834	0.9794	0.8978	0.9978
	18 h	MAE	0.0193	0.0213	0.1001	0.0116
		RMSE	0.1316	0.1470	0.3332	0.0972
		SMAPE(%)	8.51	7.54	20.84	3.93
		PCC	0.9919	0.9899	0.9534	0.9958
		R^2	0.9827	0.9784	0.8897	0.9909
Real-World	30 min	MAE	0.0112	0.0145	0.1454	0.0077
		RMSE	0.0831	0.1053	0.4702	0.0745
		SMAPE(%)	25.84	27.14	57.08	24.60
		PCC	0.9967	0.9948	0.9043	0.9974
		R^2	0.9934	0.9894	0.7850	0.9947
	60 min	MAE	0.0125	0.0139	0.1509	0.0085
		RMSE	0.0989	0.0981	0.4577	0.0700
		SMAPE(%)	27.97	27.07	61.58	28.91
		PCC	0.9953	0.9954	0.9136	0.9976
		R^2	0.9906	0.9907	0.7974	0.9952
	90 min	MAE	0.0146	0.0156	0.1520	0.0090
		RMSE	0.1187	0.1164	0.4588	0.0988
		SMAPE(%)	26.32	27.52	62.24	24.46
		PCC	0.9932	0.9935	0.9137	0.9953
		R^2	0.9863	0.9869	0.7963	0.9906

On NREL-118, AdamW achieves the lowest errors across all durations. At 6 h, 12 h, and 18 h, the MAE is 0.0077, 0.0082, and 0.0116, respectively, and the RMSE is 0.0417, 0.0478, and 0.0972. It also attains strong correlation metrics, with PCC of 0.9992, 0.9990, and 0.9958, and R^2 of 0.9984, 0.9978, and 0.9909 for 6 h, 12 h, and 18 h, respectively. In contrast, Adam yields MAE of 0.0137, 0.0143, and 0.0193 and RMSE of 0.1230, 0.1311, and 0.1316, while RMSProp yields MAE of 0.0166, 0.0180, and 0.0213 and RMSE of 0.1425, 0.1435, and 0.1470. SGD exhibits the most severe degradation, with MAE of 0.0839, 0.0943, and 0.1001 and RMSE of 0.2963, 0.3207, and 0.3332, accompanied by reduced PCC of 0.9627, 0.9563, and 0.9534 and reduced R^2 of 0.9160, 0.8978, and 0.8897.

On the real-world dataset, the same optimizer ranking holds. AdamW keeps MAE at 0.0077, 0.0085, and 0.0090 for 30 min, 60 min, and 90 min, respectively, with RMSE of 0.0745, 0.0700, and 0.0988, while maintaining PCC of 0.9974, 0.9976, and 0.9953 and R^2 of 0.9947, 0.9952, and 0.9906. Although RMSE is not strictly monotonic with duration (e.g., the 60 min case is slightly lower than the 30 min case), AdamW remains consistently better than the alternative optimizers. By contrast, SGD is highly unstable, yielding MAE of 0.1454, 0.1509, and 0.1520 and RMSE of 0.4702, 0.4577, and 0.4588, with PCC of 0.9043, 0.9136, and 0.9137 and R^2 of 0.7850, 0.7974, and 0.7963. Adam and RMSProp also underperform the default setting: Adam yields MAE of 0.0112, 0.0125, and 0.0146 with RMSE of 0.0831, 0.0989, and 0.1187, while RMSProp yields MAE of 0.0145, 0.0139, and 0.0156 with RMSE of 0.1053, 0.0981, and 0.1164. These results suggest that long contiguous gaps increase optimization difficulty and sensitivity to gradient noise, and AdamW is more robust under such training conditions, leading to more reliable optimization and superior trend preservation.

4.10. Generalization experiments

To verify the generalization capability of the proposed model, STGNI and representative baselines (BRITS, SAITS, MTC-GNN, CC-GAIN, and G-LSTM) are evaluated on (i) an unseen three-month real-grid dataset collected during the summer season (June–August 2025), which exhibits seasonal load characteristics and operating conditions different from the training period, and (ii) a perturbed NREL-118 setting created by injecting an additional 3% load fluctuation into the original data to emulate realistic operating variations. The random-missing generalization results are summarized in Table 8.

As shown in Table 8, STGNI consistently achieves the best and most stable performance in random-missing generalization experiments on both the NREL-118 and real-world datasets. In the perturbed NREL-118 setting, when the missing rate is 30%, STGNI attains an MAE of 0.0076 and an RMSE of 0.0463, which are substantially lower than those of BRITS (0.0189 and 0.0528) and SAITS (0.0215 and 0.0587). Even under 90% missingness, its MAE and RMSE remain at 0.0132 and 0.0706, respectively, while the PCC is still maintained at 0.9970, indicating strong robustness to increasing missing severity. On the real-world dataset, where non-stationary operating conditions and distribution shifts lead to markedly higher errors for all baselines, BRITS and SAITS exhibit RMSE values of 0.3380 and 0.3250 at 30% missingness, whereas STGNI still keeps the MAE and RMSE at 0.0083 and 0.0825 with a PCC of 0.9966. Even at 90% missingness, its RMSE remains significantly lower than that of G-LSTM (0.3350) and CC-GAIN (0.4280). This performance advantage can be attributed to the joint design of missing-aware mechanisms, spatial topological constraints, and temporal modeling in STGNI, which enables effective exploitation of structural redundancy and temporal

Table 8
Generalization performance under random missing patterns.

Dataset	Missing	Metric	BRITS	SAITS	MTC-GNN	CC-GAIN	G-LSTM	STGNI
NREL-118	Random (30%)	MAE	0.0189	0.0215	0.0608	0.0210	0.0395	**0.0076**
		RMSE	0.0528	0.0587	0.2685	0.0580	0.1365	**0.0463**
		SMAPE(%)	8.10	9.65	21.20	9.25	19.50	**3.65**
		PCC	0.9982	0.9978	0.9605	0.9979	0.9902	**0.9989**
		R^2	0.9965	0.9957	0.9150	0.9959	0.9815	**0.9976**
	Random (60%)	MAE	0.0236	0.0268	0.0789	0.0261	0.0458	**0.0096**
		RMSE	0.0639	0.0708	0.3052	0.0695	0.1512	**0.0558**
		SMAPE(%)	9.55	11.10	26.80	10.80	23.60	**4.25**
		PCC	0.9973	0.9968	0.9490	0.9969	0.9875	**0.9982**
		R^2	0.9950	0.9940	0.8855	0.9942	0.9770	**0.9965**
	Random (90%)	MAE	0.0309	0.0345	0.0965	0.0338	0.0537	**0.0132**
		RMSE	0.0815	0.0899	0.3480	0.0885	0.1706	**0.0706**
		SMAPE(%)	12.60	14.10	33.40	13.50	28.40	**5.60**
		PCC	0.9959	0.9952	0.9305	0.9954	0.9820	**0.9970**
		R^2	0.9926	0.9911	0.8450	0.9918	0.9690	**0.9941**
Real-world	Random (30%)	MAE	0.0820	0.0615	0.0845	0.0640	0.0510	**0.0083**
		RMSE	0.3380	0.3250	0.3180	0.3320	0.2650	**0.0825**
		SMAPE(%)	52.0	46.5	26.5	47.8	41.0	**28.3**
		PCC	0.9440	0.9505	0.9445	0.9480	0.9685	**0.9966**
		R^2	0.8850	0.8955	0.8855	0.8920	0.9370	**0.9925**
	Random (60%)	MAE	0.0915	0.0728	0.0980	0.0758	0.0578	**0.0092**
		RMSE	0.3720	0.3580	0.3550	0.3660	0.2880	**0.0968**
		SMAPE(%)	58.0	52.5	30.5	53.8	45.0	**31.6**
		PCC	0.9300	0.9385	0.9285	0.9350	0.9600	**0.9950**
		R^2	0.8580	0.8730	0.8550	0.8680	0.9200	**0.9890**
	Random (90%)	MAE	0.1120	0.0930	0.1220	0.0965	0.0710	**0.0128**
		RMSE	0.4470	0.4210	0.4380	0.4280	0.3350	**0.1320**
		SMAPE(%)	69.5	63.5	39.0	64.0	56.0	**36.8**
		PCC	0.9050	0.9160	0.9020	0.9120	0.9430	**0.9920**
		R^2	0.8000	0.8220	0.7920	0.8160	0.8850	**0.9820**

Table 9
RMSE improvement (%) of state estimation results obtained using STGNI-imputed data over MTC-GNN-imputed data under different training ratios.

Model (Improvement of	60%		70%		80%	
STGNI over MTC-GNN)	S_{V_m}	S_{V_a}	S_{V_m}	S_{V_a}	S_{V_m}	S_{V_a}
WLS	1.6%	1.3%	1.3%	1.1%	1.0%	0.8%
GCLSE	1.5%	1.2%	1.2%	1.0%	0.9%	0.7%
PINN	1.8%	1.5%	1.4%	1.2%	1.1%	0.9%
Transformer-Bi-LSTM	8.6%	8.1%	7.1%	6.6%	6.0%	5.6%

consistency under high missing rates and distribution shifts. In contrast, purely temporal methods such as BRITS and SAITS suffer from pronounced error accumulation as missingness increases, while MTC-GNN, despite achieving relatively lower SMAPE values, exhibits inferior MAE, RMSE, and PCC (e.g., RMSE of 0.3550 and PCC of 0.9285 at 60% missingness), suggesting that relying solely on spatial graph modeling is insufficient to simultaneously guarantee magnitude accuracy and global correlation under complex real-world operating conditions.

4.11. State estimation impact experiments

State estimation aims to infer the system operating state, including bus voltage magnitudes V_m and phase angles V_a, from imperfect measurements. Since missing and noisy data can substantially degrade estimation accuracy, improving measurement quality through effective imputation is critical for reliable downstream analysis.

To assess the impact of imputation quality on state estimation, four representative estimators—Transformer-Bi-LSTM [8], GCLSE [38], WLS [39], and PINN [40]—are evaluated using measurements imputed by MTC-GNN and STGNI under three training-set proportions (60%, 70%, and 80%). All experiments are conducted on Real-World data, and RMSE is used as the evaluation metric.

As shown in Table 9, STGNI-imputed data consistently reduce the RMSE of all state estimation models compared with MTC-GNN. The largest improvements are observed for Transformer-Bi-LSTM, with RMSE reductions of approximately 6–9%, indicating its strong sensitivity to input data quality. In contrast, WLS exhibits smaller but stable gains (about 0.8–1.6%), reflecting the robustness of physics-based optimization. GCLSE and PINN achieve moderate yet consistent improvements (about 0.7–1.8%), as their graph priors and physical constraints partially alleviate error propagation. As the training proportion increases, the improvement margins gradually decrease, suggesting reduced reliance on imputation quality when more labeled data are available. Overall, these results confirm that higher-quality imputation directly translates into more accurate and stable state estimation, further validating the effectiveness of STGNI.

5. Conclusion

This study presents STGNI, a spatio-temporal graph neural network for recovering missing measurements in power-system state estimation. The model handles both random and continuous missingness commonly encountered in SCADA environments. In practice, generator-side measurements are typically reliable, whereas transmission-line flow measurements are more susceptible to communication interruptions, device failures, and environmental disturbances. Accordingly, we focus on imputing line-related features to better reflect real-world engineering scenarios. By integrating missing-aware encoding, dual-channel spatial structure modeling, and hybrid short–long-term temporal modeling into a unified framework, STGNI captures complex spatial correlations and temporal dynamics without assuming a specific missing mechanism, enabling accurate data recovery.

Extensive experiments on the NREL-118 benchmark system and a real-world power-grid dataset yield the following findings: (i) STGNI consistently outperforms state-of-the-art imputation baselines under diverse missing configurations, exhibiting strong robustness from low to

high missing rates and under different data degradations. (ii) Improved imputation quality translates into better downstream state estimation. Using STGNI-recovered measurements yields lower RMSE across multiple state estimation models, including WLS, GCLSE, PINN, and Transformer-BiLSTM, further validating the practical value of STGNI for missing-measurement recovery.

Overall, STGNI effectively enhances data quality in modern power systems and improves the reliability of downstream estimation and monitoring applications. Future work will explore: (i) extending STGNI to multi-modal measurement settings (e.g., PMU–SCADA fusion); (ii) assessing cross-system generalization to different grid topologies and load profiles without full retraining.

CRediT authorship contribution statement

Ping Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Zhen Dai:** Writing – review & editing, Supervision, Resources, Formal analysis. **Zhenyu Wang:** Writing – review & editing, Supervision, Resources, Formal analysis. **Yuyuan Yu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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